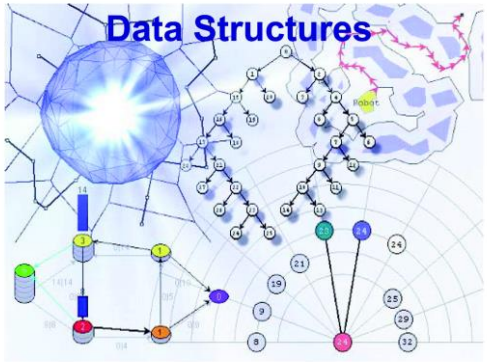
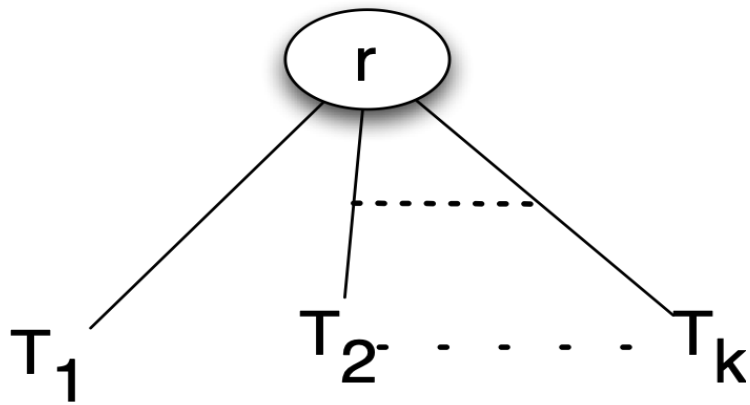
BBM 201
DATA STRUCTURES

Lecture 7:
Tree, Binary Trees and Binary Search Trees



What is a Tree?

- T is a **tree** if either
  - T has no nodes, or
  - T is of the form:



where r is a node and T_1, T_2, ..., T_k are trees.

Tree Terminology

Parent – The parent of node n is the node directly above in the tree.

Child – The child of node n is the node directly below in the tree.

- If node m is the parent of node n, node n is the child of node m.

Root – The only node in the tree with no parent.

Leaf – A node with no children.

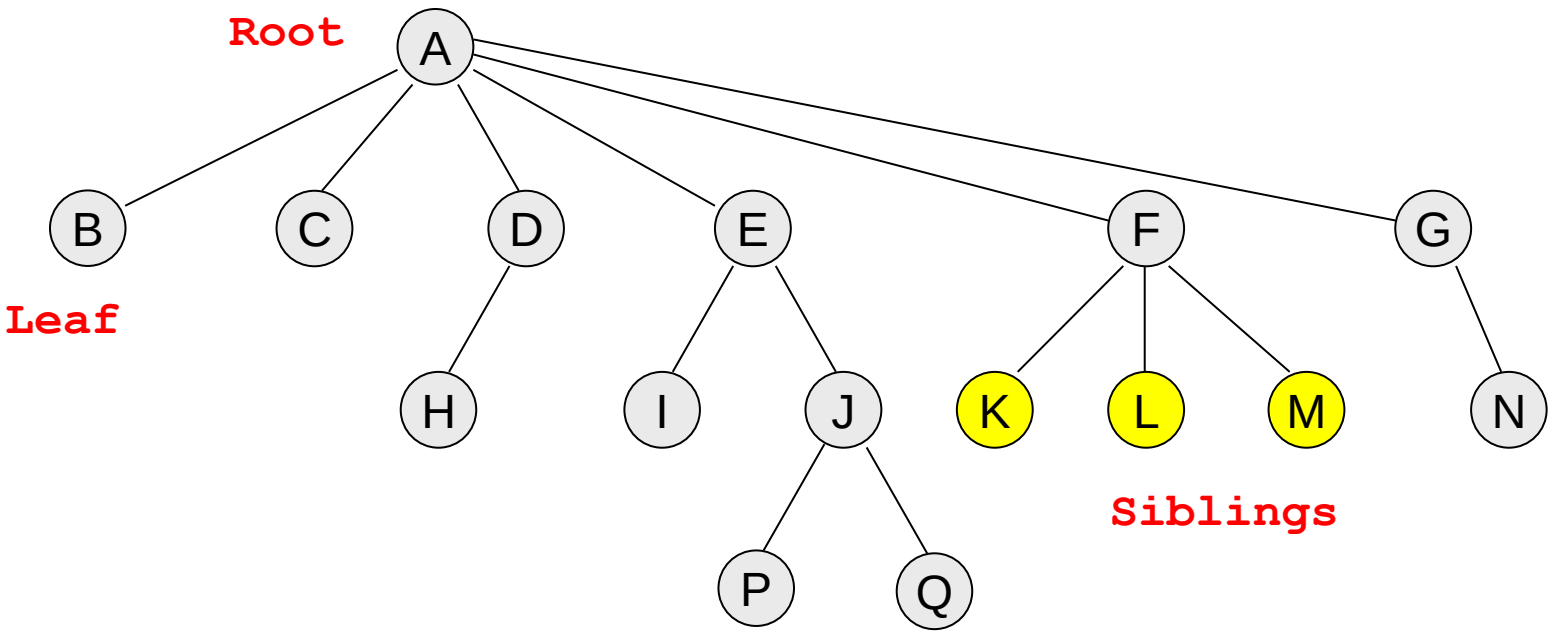
Siblings – Nodes with a common parent.

Ancestor – An ancestor of node n is a node on the path from the root to n.

Descendant – A descendant of node n is a node on the path from n to a leaf.

Subtree – A subtree of node n is a tree that consists of a child (if any) of n and the child's descendants (a tree which is rooted by a child of node n)

A Tree – Example



–Node *A* has 6 ***children***: B, C, D, E, F, G. These nodes are ***siblings***.

–B, C, H, I, P, Q, K, L, M, N are ***leaves*** in the tree above.

–K, L, M are ***siblings*** since F is parent of all of them. I, J and P, Q are the other siblings…

–E is an ***ancestor*** of P, Q, I and J.

–H is a ***descendant*** of A and D.

What is a Tree?

- The root of each sub-tree is said to be ***child*** of r, and r is the ***parent*** of each sub-tree's root.

- If a tree is a collection of N nodes, then it has N-1 edges. **Why?**

- A ***path*** from node n_1 to n_k is defined as a sequence of nodes $n_1, n_2, \ldots, n_k$ such that n_i is parent of n_{i+1} $(1 \leq i < k)$
  - There is a path from every node to itself.
  - There is exactly one path from the root to each node. **Why?**

Level of a node

Level – The level of node n is the number of nodes on the path from root to node n.

Definition: *The level of node n in a tree T*

- If n is the root of T, the level of n is 1.
- If n is not the root of T, its level is 1 greater than the level of its parent.

Level of a node

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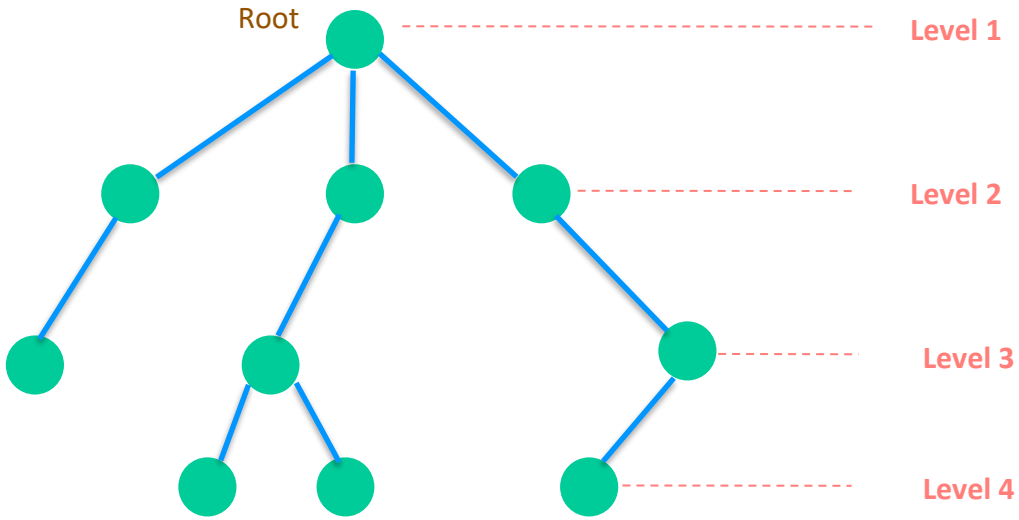
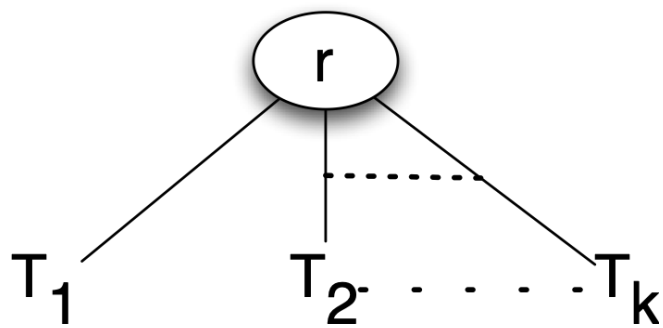


Height of A Tree

Height – number of nodes on **longest path** from the root to any leaf.

- The height of a tree T in terms of the levels of its nodes is defined as:
  - If T is empty, its height is 0
  - If T is not empty, its height is equal to the maximum level of its nodes.

- Or, the height of a tree T can be defined recursively as:
  - If T is empty, its height is 0.
  - If T is non-empty tree, then since T is of the form:

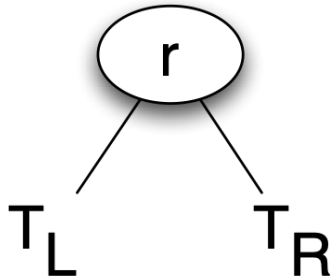


$$\textbf{height(T)} = \textbf{1} + \textbf{max}\{height(T_1), height(T_2), \ldots, height(T_k)\}$$

Binary Tree

- *A binary tree T is a set of nodes with the following properties:*
  - *The set can be empty.*
  - *Otherwise, the set is partitioned into three disjoint subsets:*
    - *a tree consists of a distinguished node r, called **root**, and*
    - *two possibly empty sets are binary tree, called **left** and **right subtrees** of r.*

- T is a **binary tree** if either
  - T has no nodes, or
  - T is of the form:



where r is a node and T_L and T_R are binary trees.

Binary Tree Terminology

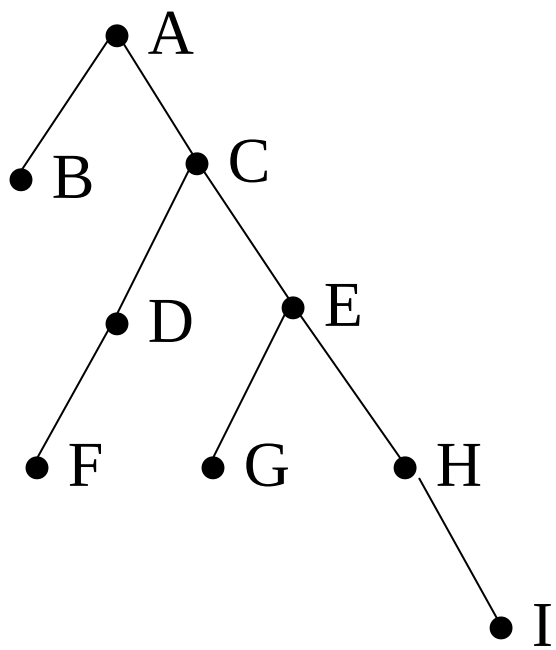
Left Child – The left child of node n is a node directly below and to the left of node n in a binary tree.

Right Child – The right child of node n is a node directly below and to the right of node n in a binary tree.

Left Subtree – In a binary tree, the left subtree of node n is the left child (if any) of node n plus its descendants.

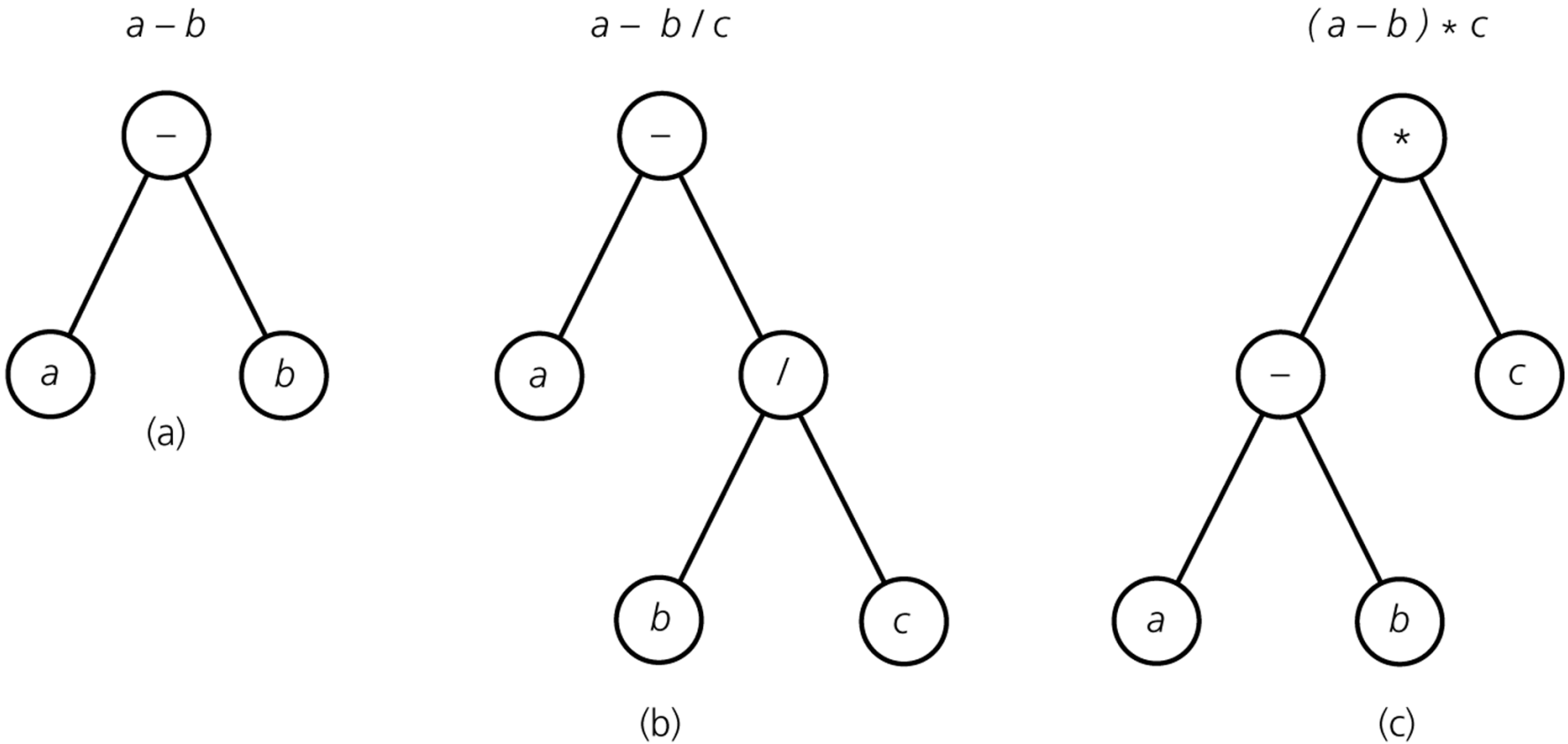
Right Subtree – In a binary tree, the right subtree of node n is the right child (if any) of node n plus its descendants.

Binary Tree -- Example



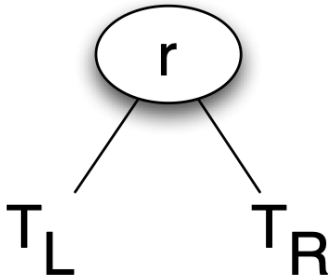
- A is the root.
- B is left child of A,
  C is right child of A.
- D doesn't have a right child.
- H doesn't have a left child.
- B, F, G and I are leaves.

Binary Tree – Representing Algebraic Expressions



Height of Binary Tree

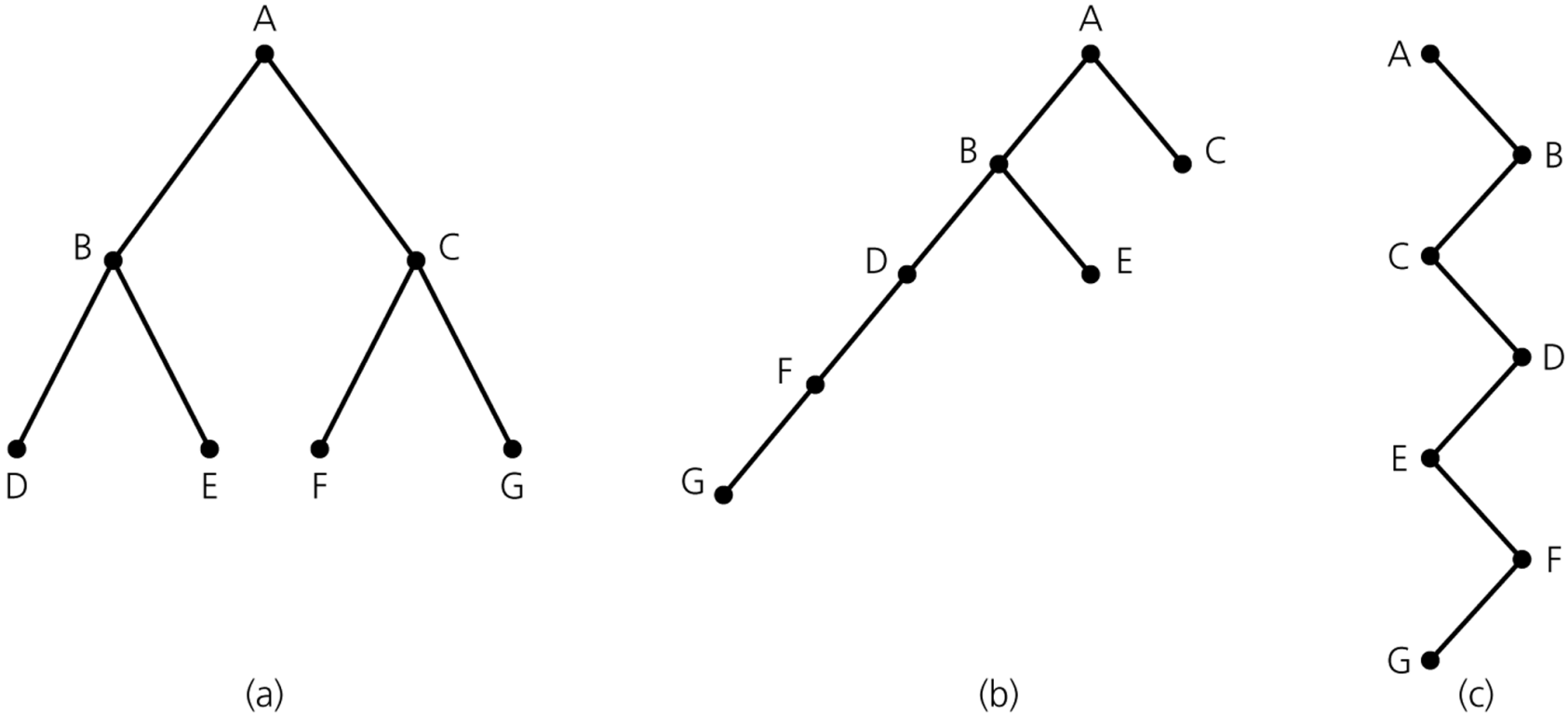
- The height of a binary tree T can be defined as recursively as:
  - If T is empty, its height is 0.
  - If T is non-empty tree, then since T is of the form ...



... height of T is 1 greater than height of its root's taller subtree; ie.

$$\textbf{height(T)} = 1 + \textbf{max}\{\text{height}(T_L),\text{height}(T_R)\}$$

Height of Binary Tree (cont.)

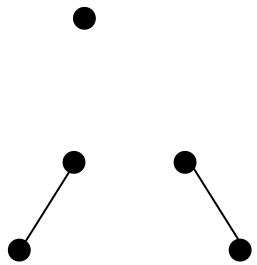


Binary trees with the same nodes but different heights

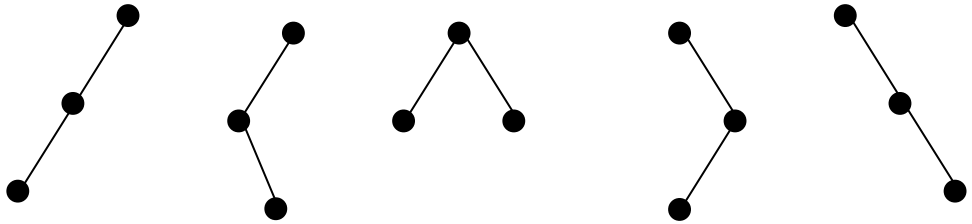
Number of Structurally Different Binary trees with n Nodes

n=0 empty tree (1 tree)

n=1 (1 tree)



n=2 (2 trees)



n=3 (5 trees)

To express, *structurally different* binary trees with n nodes:
Let us arbitrarily order the elements in the tree from 1, 2 … , n, and assume that only the lower order nodes can be placed as the left child to the root node and higher order nodes as the right child.

Number of Structurally Different Binary trees with n Nodes

To express, *structurally different* binary trees with n nodes:
Let us arbitrarily order the elements in the tree from 1, 2 … , n, and assume that only the lower order nodes can be placed as the left child to the root node and higher order nodes as the right child.

We can choose root n different ways:

- choose node 1 as the root; all the remaining (n-1) elements can be inserted as the right subtree
- choose node 2 as the root; 1 element can be inserted as the left subtree, and (n-2) elements can be inserted as the right subtree
- choose node 3 as the root node; 2 elements can be inserted as the left subtree and (n-3) elements can be inserted as the right subtree

…

- choose node i as the root node; (i - 1) elements can be inserted as left subtree and (n-i) elements can be inserted as the right subtree.

Number of Structurally Different Binary trees with n Nodes

To express, *structurally different* binary trees with n nodes:
Let us arbitrarily order the elements in the tree from 1, 2 … , n, and assume that only the lower order nodes can be placed as the left child to the root node and higher order nodes as the right child.

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…

- choose node i as the root node; (i - 1) elements can be inserted as left subtree and (n-i) elements can be inserted as the right subtree.

All subtrees are also binary trees. So, the number of binary trees can be computed using a recursive function f(n), where the base cases **`f(0)=1`** (only one way to generate empty binary tree) and **`f(1)=1`** (1 way to generate a BT with one node) as follows:

`f(n) = f(0)f(n-1) + f(1)f(n-2) + … + f(i-1)f(n-i) + … + f(n-1)f(0)`

$$f(n) = \sum_{i=1}^{n} f(i-1)f(n-i)$$

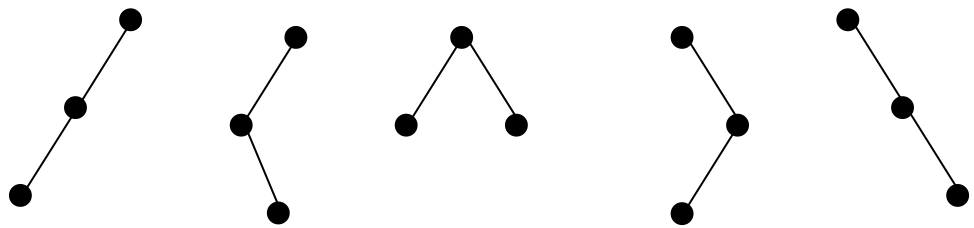
```
f(0) = 1
f(1) = 1
f(2) = 2
f(3) = 5
f(4) = ?
```

Number of Structurally Different Binary trees with n Nodes

n=0 ❾ empty tree (1 tree)

n=1 ❾ (1 tree)

n=2 ❾ (2 trees)

n=3 ❾  (5 trees)

n is even ❾

$$numBT(n) = 2 \sum_{i=0}^{(n-1)/2} (numBT(i) \times numBT(n-i-1))$$

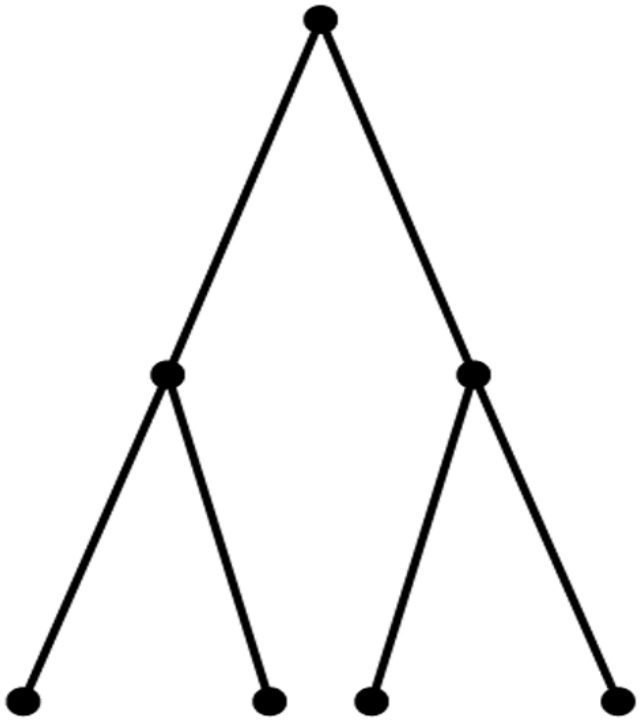
n is odd ❾

$$numBT(n) = 2 \sum_{i=0}^{(n-1)/2)-1} (numBT(i) \times numBT(n-i-1)) + numBT((n-1)/2) \times numBT((n-1)/2)$$

Full Binary Tree

- In a ***full binary tree*** of height h, all nodes that are at a level less than h have two children each.
- Each node in a full binary tree has left and right subtrees of the same height.
- Among binary trees of height h, a full binary tree has as many leaves as possible, and **all the leaves are at level h**.
- A full binary tree has **no missing nodes**.
- Recursive definition of full binary tree:
  - If T is empty, T is a full binary tree of height 0.
  - If T is not empty and has height h>0, T is a full binary tree if its root's subtrees are both full binary trees of height h-1.

Full Binary Tree – Example



A full binary tree of height 3

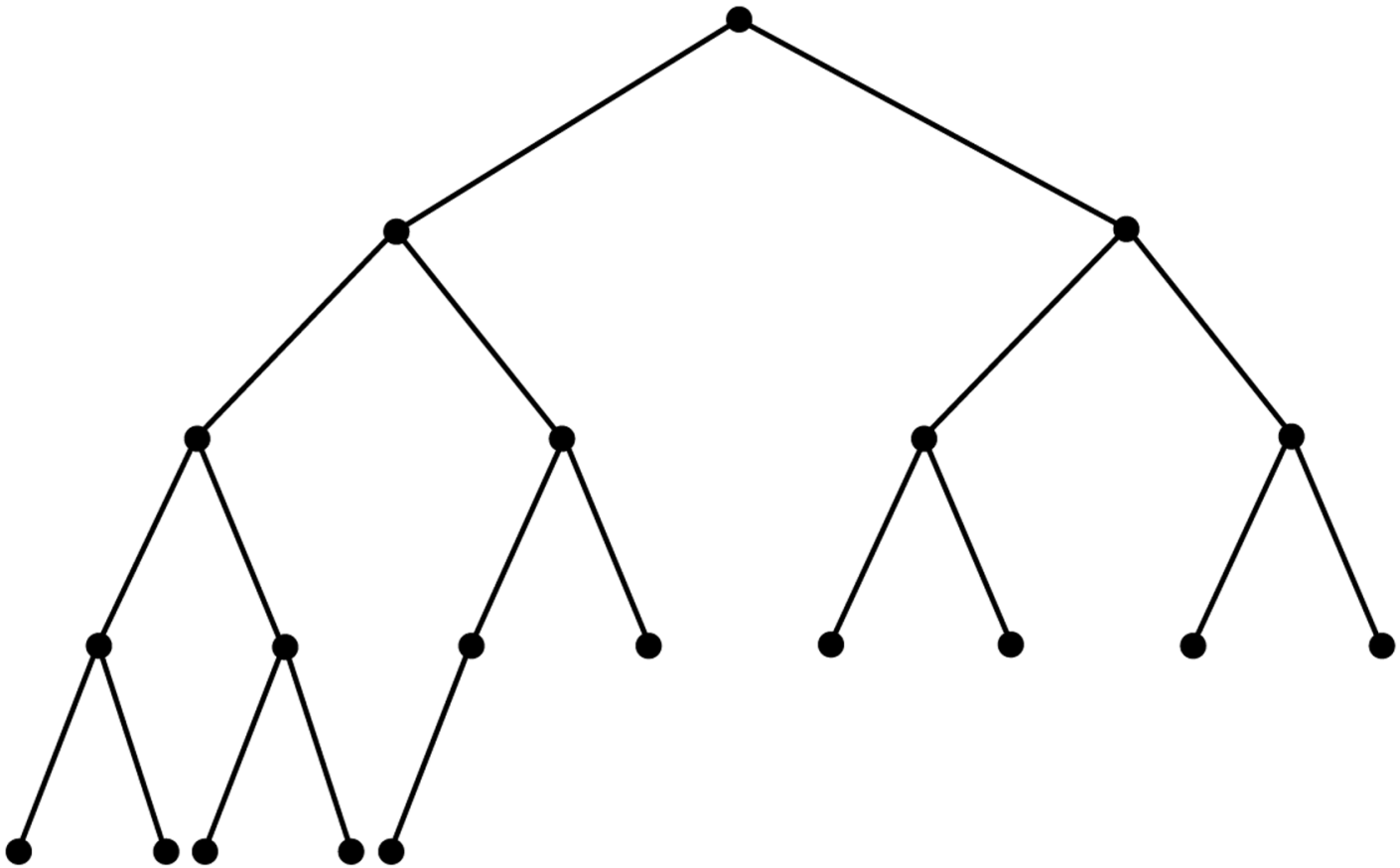
Complete Binary Tree

- A ***complete binary tree*** of height ***h*** is a binary tree that is **full down to level h-1**, with level h filled in from left to right.

- A binary tree T of height h is complete if
  1. All nodes at level h-2 and above have two children each, and
  2. When a node at level h-1 has children, all nodes to its left at the same level have two children each, and
  3. When a node at level h-1 has one child, it is a left child.

- A full binary tree is a complete binary tree.

Complete Binary Tree – Example



Balanced Binary Tree

- A binary tree is ***balanced*** (or ***height balanced***), if the height of any node's right subtree and left subtree **differ no more than 1**.

- A complete binary tree is a balanced tree. Why?

- Later, we look at other height balanced trees.
  - AVL trees
  - Red-Black trees, ...

Maximum and Minimum Heights of a Binary Tree

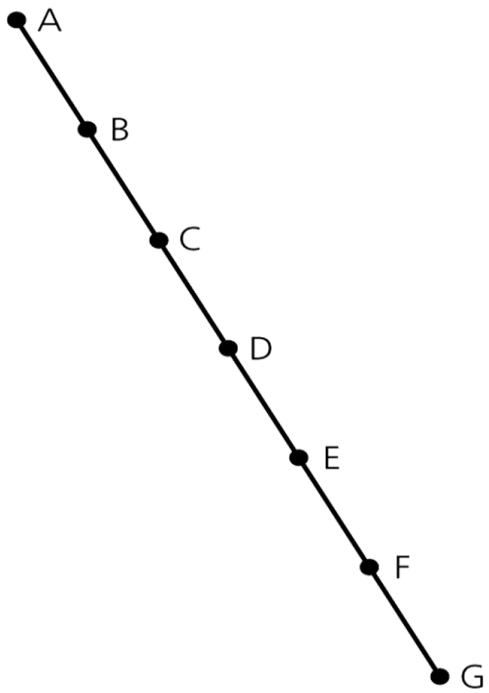
- **Efficiency** of most binary tree operations **depends on tree height**.

- **E.g. maximum number of key comparisons** for retrieval, deletion, and insertion operations for BSTs is the height of the tree.

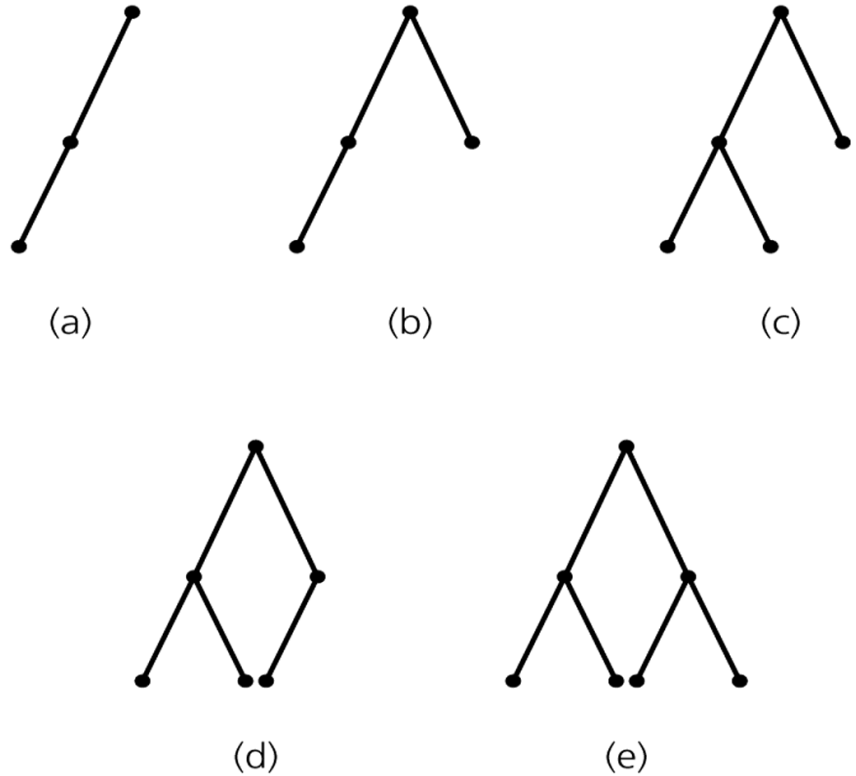
- The maximum height of a binary tree with n nodes is n. How?

- Each level of a minimum height tree, except the last level, must contain as many nodes as possible.
  - Should the tree be a Complete Binary Tree?

Maximum and Minimum Heights of a Binary Tree

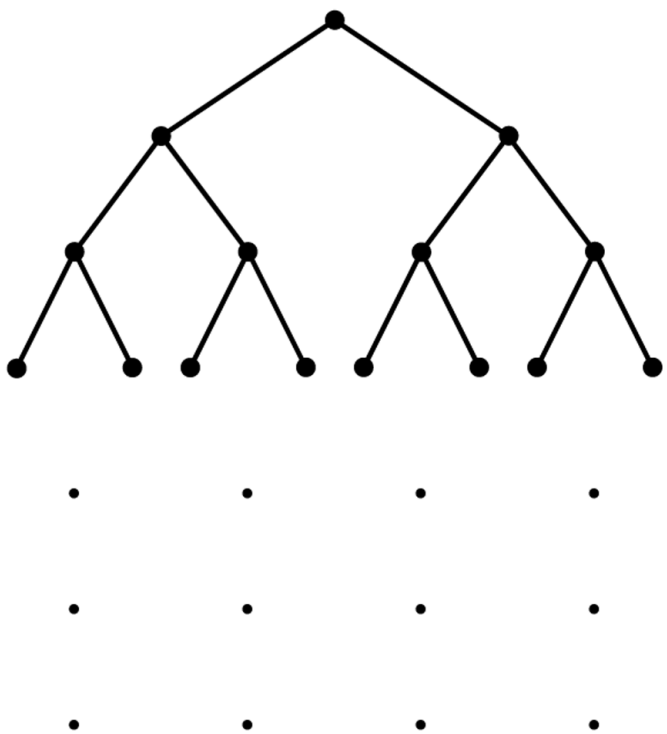


A maximum-height binary tree with seven nodes



Some binary trees of height 3

Counting the nodes in a full binary tree of height *h*



Level	Number of nodes at this level	Number of nodes at this and previous levels
1	$1 = 2^0$	$1 = 2^1 - 1$
2	$2 = 2^1$	$3 = 2^2 - 1$
3	$4 = 2^2$	$7 = 2^3 - 1$
4	$8 = 2^3$	$15 = 2^4 - 1$
.	.	.
.	.	.
.	.	.
h	2^{h-1}	$2^h - 1$

Some Height Theorems

Theorem: A full binary tree of height $h \geq 0$ has 2^h-1 nodes.

- The maximum number of nodes that a binary tree of height h can have is 2^h-1.

- We cannot insert a new node into a full binary tree without increasing its height.

Some Height Theorems

Theorem 10-4: The minimum height of a binary tree with n nodes is $\lceil \log_2(n+1) \rceil$.

Proof: Let h be the smallest integer such that $n \leq 2^h-1$. We can establish following facts:

Fact 1 – A binary tree whose height is $\leq$ h-1 has $<$ n nodes.

- Otherwise h cannot be smallest integer in our assumption.

Fact 2 – There exists a complete binary tree of height h that has exactly n nodes.

- A full binary tree of height h-1 has $2^{h-1}-1$ nodes.
- Since a binary tree of height h cannot have more than 2^h-1 nodes.
- At level h, we will reach n nodes.

Fact 3 – The minimum height of a binary tree with n nodes is the smallest integer h such that $n \leq 2^h-1$.

So, ➔ $2^{h-1}-1 < n \leq 2^h-1$

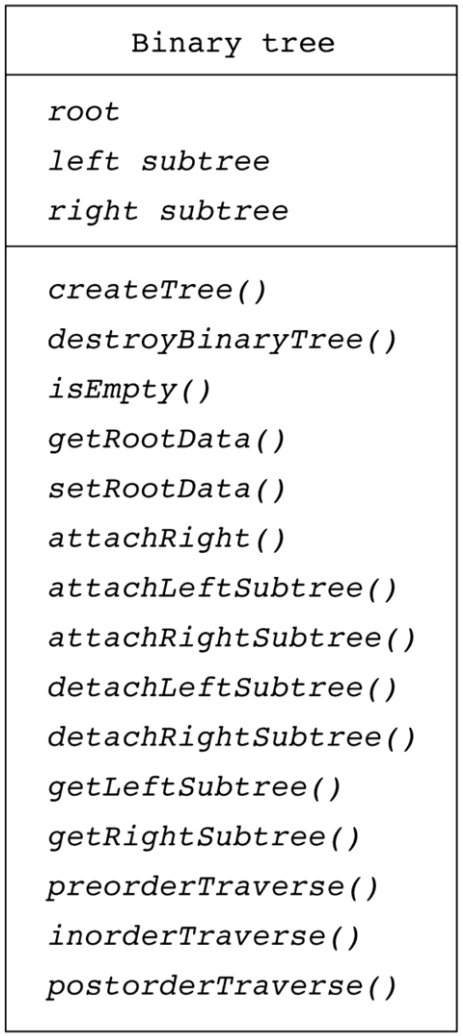
➔ $2^{h-1} < n+1 \leq 2^h$

➔ $h-1 < \log_2(n+1) \leq h$

Thus, ➔ $h = \lceil \log_2(n+1) \rceil$ is the minimum height of a binary tree with n nodes.

- UML Diagram for **BinaryTree ADT**

- What is an **ADT**?

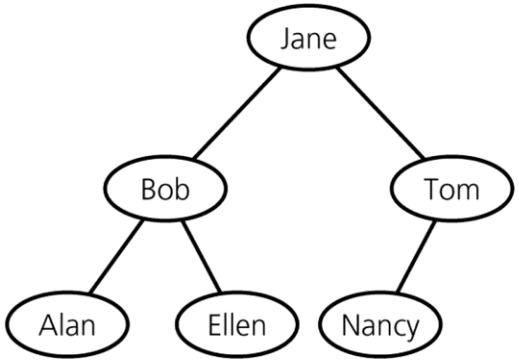


An Array-Based Implementation of Binary Trees

```
const int MAX_NODES = 100;              // maximum number of nodes
typedef string TreeItemType;

class TreeNode {                        // node in the tree
private:
        TreeNode();
        TreeNode(const TreeItemType& nodeItem, int left, int right);

        TreeItemType item;              // data portion
        int leftChild;                  // index to left child
        int rightChild;                 // index to right child

        // friend class - can access private parts
        friend class BinaryTree;
};

// An array of tree nodes
TreeNode[MAX_NODES] tree;
int  root;
int  free;
```

An Array-Based Implementation (cont.)

(a)



(b)

tree

	item	leftChild	rightChild
0	Jane	1	2
1	Bob	3	4
2	Tom	5	–1
3	Alan	–1	–1
4	Ellen	–1	–1
5	Nancy	–1	–1
6	?	–1	7
7	?	–1	8
8	?	–1	9
.	.	.	.
.	.	.	.
.	.	.	.

root: 0

free: 6

Free list

- A ***free list*** keeps track of available nodes.
- To insert a new node into the tree, we first obtain an available node from the free list.
- When we delete a node from the tree, we have to place into the free list so that we can use it later.

An Array-Based Representation of a Complete Binary Tree

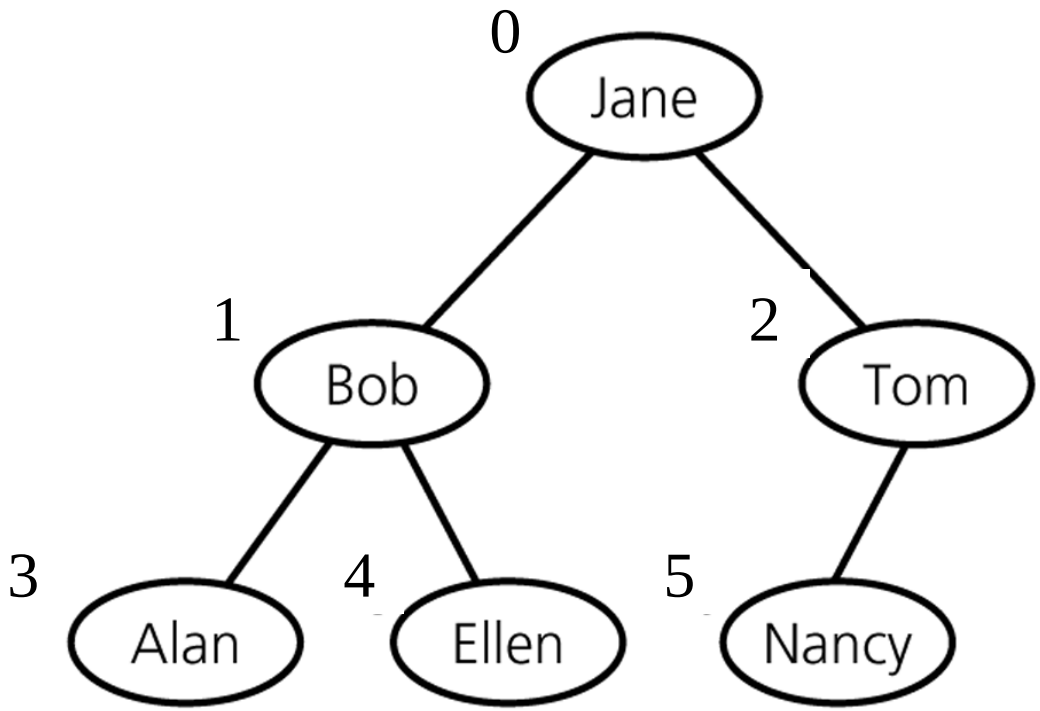
- If we know that our binary tree is a **complete binary tree**, we can use a simpler array-based representation for complete binary trees
  - **without** using **leftChild, rightChild** links

- We can number the nodes level by level, and left to right (starting from 0, the root will be 0). If a node is numbered as i, in the ith location of the array, `tree[i]`, contains this node without links.

- Using these numbers we can find leftChild, rightChild, and parent of a node i.

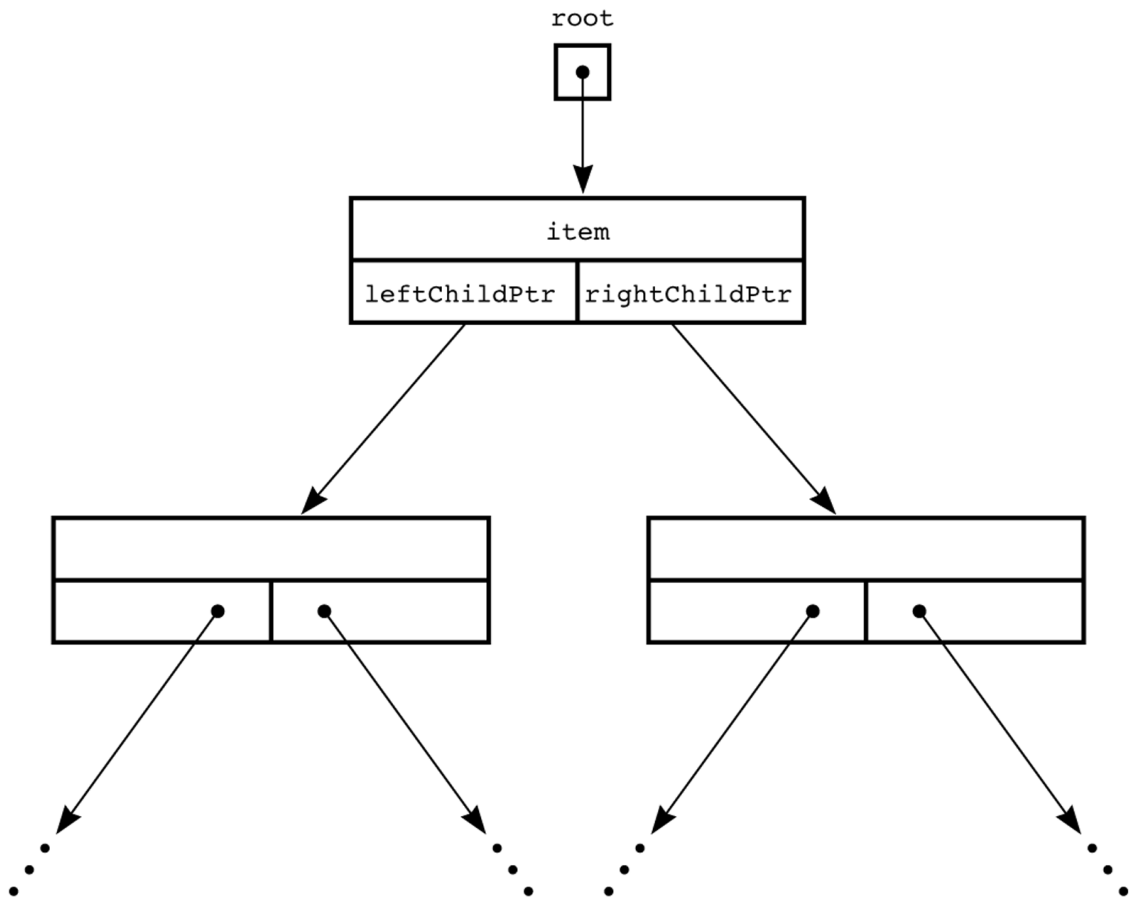
The left child (if it exists) of node i is	`tree[2*i+1]`
The right child (if it exists) of node i is	`tree[2*i+2]`
The parent (if it exists) of node i is	`tree[(i-1)/2]`

An Array-Based Representation of a Complete Binary Tree (cont.)



0	Jane
1	Bob
2	Tom
3	Alan
4	Ellen
5	Nancy
6	
7	

Pointer-Based Implementation of Binary Trees



A Pointer-Based Implementation of a Binary Tree Node

```
typedef string TreeItemType;

class TreeNode {          // node in the tree
private:
   TreeNode() {}
   TreeNode(const TreeItemType& nodeItem,
      TreeNode *left = NULL,
      TreeNode *right = NULL)
      :item(nodeItem),leftChildPtr(left),rightChildPtr(right) {}

   TreeItemType item;      // data portion
   TreeNode *leftChildPtr;  // pointer to left child
   TreeNode *rightChildPtr; // pointer to right child

   friend class BinaryTree;
};
```

Binary Tree – TreeException.h

```
class TreeException : public exception{

private:
   string msg;

public:
        virtual const char* what() const throw()
        {
             return msg.c_str();
        }
  TreeException(const string & message =""):
        exception(), msg(message) {};
  ~TreeException() throw() {};

}; // end TreeException
```

Binary tree
root *left subtree* *right subtree*
createTree() *destroyBinaryTree()* *isEmpty()* *getRootData()* *setRootData()* *attachRight()* *attachLeftSubtree()* *attachRightSubtree()* *detachLeftSubtree()* *detachRightSubtree()* *getLeftSubtree()* *getRightSubtree()* *preorderTraverse()* *inorderTraverse()* *postorderTraverse()*

The BinaryTree Class

- **Properties**
  - TreeNode * root

- **Constructors**
  - BinaryTree();
  - BinaryTree(const TreeItemType& rootItem);
  - BinaryTree(const TreeItemType& rootItem,
    BinaryTree& leftTree, BinaryTree& rightTree);
  - BinaryTree(const BinaryTree& tree);
    void copyTree(TreeNode *treePtr, TreeNode* & newTreePtr) const;

- **Destructor**
  - ~BinaryTree();
    void destroyTree(TreeNode * &treePtr);

BinaryTree: Public Methods

- bool isEmpty()
- TreeItemType rootData() const throw(TreeException)
- void setRootData(const TreeItemType& newItem)
- void attachLeft(const TreeItemType& newItem)
- void attachRight(const TreeItemType& newItem)
- void attachLeftSubtree(BinaryTree& leftTree)
- void attachRightSubtree(BinaryTree& rightTree)
- void detachLeftSubtree(BinaryTree& leftTree)
- void detachRightSubtree(BinaryTree& rightTree)
- BinaryTree leftSubtree()
- BinaryTree rightSubtree()
- void preorderTraverse(FunctionType visit_fn)
- void inorderTraverse(FunctionType visit_fn)
- void postorderTraverse(FunctionType visit_fn)
  - FunctionType is a pointer to a function:
    - typedef void (*FunctionType)(TreeItemType& anItem);

BinaryTree: Implementation

- The complete implementation is in
  - Frank M. Carrano and Timothy Henry, *Data Abstraction and Problem Solving with C++: Walls and Mirrors*, 7th edition, Pearson, 2017

- In class, we will go through only some methods
  - Skipping straightforward methods
    - Such as `isEmpty`, `rootData`, and `setRootData` functions
  - Skipping some details
    - Such as throwing exceptions

```
// Default constructor
BinaryTree::BinaryTree() : root(NULL) {

}


// Protected constructor
BinaryTree::BinaryTree(TreeNode *nodePtr) : root(nodePtr) {

}


// Constructor
BinaryTree::BinaryTree(const TreeItemType& rootItem) {
        root = new TreeNode(rootItem, NULL, NULL);
}
```

```
// Constructor
BinaryTree::BinaryTree(const TreeItemType& rootItem,
                                        BinaryTree& leftTree, BinaryTree& rightTree) {
        root = new TreeNode(rootItem, NULL, NULL);
        attachLeftSubtree(leftTree);
        attachRightSubtree(rightTree);
}

void BinaryTree::attachLeftSubtree(BinaryTree& leftTree) {
        // Assertion: nonempty tree; no left child
        if (!isEmpty() && (root->leftChildPtr == NULL)) {
                root->leftChildPtr = leftTree.root;
                leftTree.root = NULL
        }
}

void BinaryTree::attachRightSubtree(BinaryTree& rightTree) {
        // Left as an exercise
}
```

```
// Copy constructor
BinaryTree::BinaryTree(const BinaryTree& tree) {
        copyTree(tree.root, root);
}


// Uses preorder traversal for the copy operation
// (Visits first the node and then the left and right children)
void BinaryTree::copyTree(TreeNode *treePtr, TreeNode *& newTreePtr) const {

        if (treePtr != NULL) {          // copy node
                newTreePtr = new TreeNode(treePtr->item, NULL, NULL);
                copyTree(treePtr->leftChildPtr, newTreePtr->leftChildPtr);
                copyTree(treePtr->rightChildPtr, newTreePtr->rightChildPtr);
        }
        else
                newTreePtr = NULL;   // copy empty tree
}
```

```
// Destructor
BinaryTree::~BinaryTree() {
        destroyTree(root);
}


// Uses postorder traversal for the destroy operation
// (Visits first the left and right children and then the node)
void BinaryTree::destroyTree(TreeNode *& treePtr) {

        if (treePtr != NULL){
                destroyTree(treePtr->leftChildPtr);
                destroyTree(treePtr->rightChildPtr);
                delete treePtr;
                treePtr = NULL;
        }
}
```

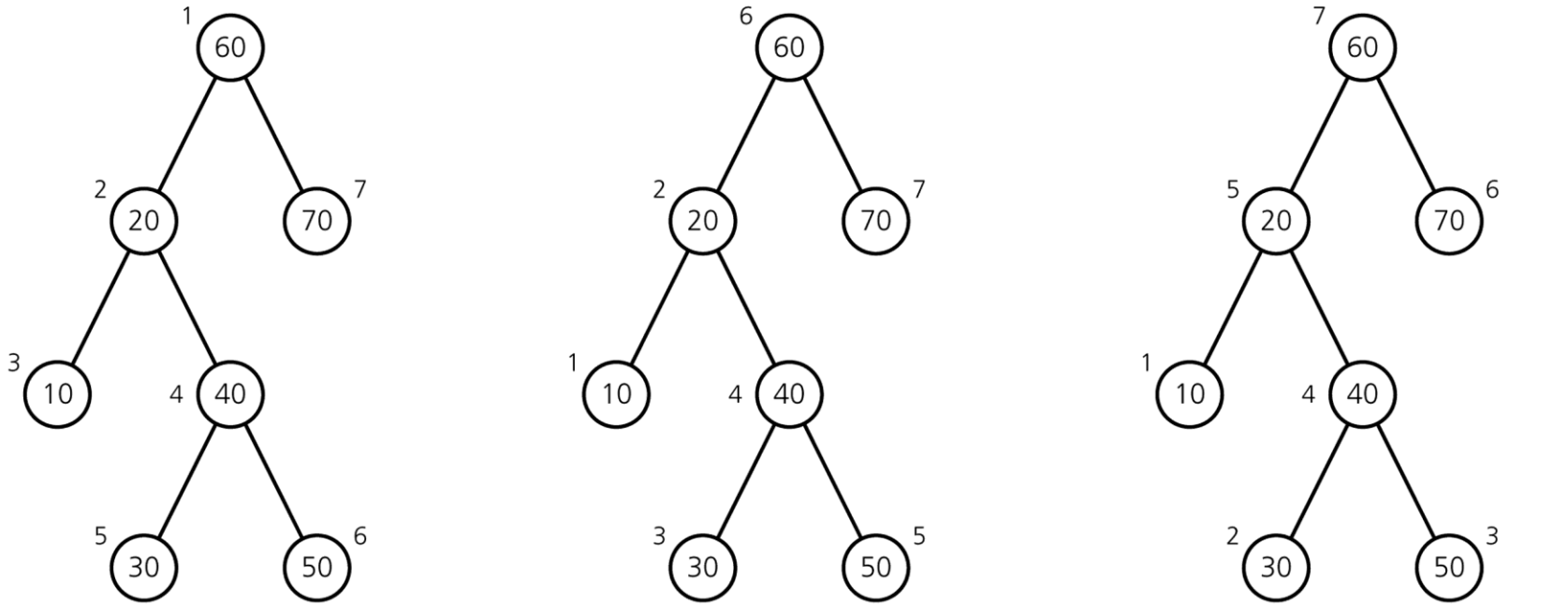
Binary Tree Traversals

- **Preorder Traversal**
  - The node is visited before its left and right subtrees,
  - Visit the node, visit the left subtree, visit the right subtree

- **Postorder Traversal**
  - The node is visited after both subtrees.
  - Visit the left subtree, visit the right subtree, visit the node

- **Inorder Traversal**
  - The node is visited between the subtrees,
  - Visit the left subtree, visit the node, and visit the right subtree.

Binary Tree Traversals



(a) Preorder: 60, 20, 10, 40, 30, 50, 70

(b) Inorder: 10, 20, 30, 40, 50, 60, 70

(c) Postorder: 10, 30, 50, 40, 20, 70, 60

(Numbers beside nodes indicate traversal order.)

```
void BinaryTree::preorderTraverse(FunctionType visit) {
        preorder(root, visit);
}
void BinaryTree::inorderTraverse(FunctionType visit) {
        inorder(root, visit);
}
void BinaryTree::postorderTraverse(FunctionType visit) {
        postorder(root, visit);
}
```

Remember that:

FunctionType is a pointer to a function

- Variables that point to the address of a function
- `typedef void (*FunctionType)(TreeItemType& anItem);`

Example of using inorderTraverse function:

- `void display(TreeItemType& anItem) { cout << anItem << endl; }`
- `BinaryTree T1;`

  `T1.inorderTraverse(display);`

```
void BinaryTree::preorder(TreeNode *treePtr, FunctionType visit) {
        if (treePtr != NULL) {
                visit(treePtr->item);
                preorder(treePtr->leftChildPtr, visit);
                preorder(treePtr->rightChildPtr, visit);
        }
}

void BinaryTree::inorder(TreeNode *treePtr, FunctionType visit) {
        if (treePtr != NULL) {
                inorder(treePtr->leftChildPtr, visit);
                visit(treePtr->item);
                inorder(treePtr->rightChildPtr, visit);
        }
}

void BinaryTree::postorder(TreeNode *treePtr, FunctionType visit) {
        if (treePtr != NULL) {
                postorder(treePtr->leftChildPtr, visit);
                postorder(treePtr->rightChildPtr, visit);
                visit(treePtr->item);
        }
}
```

Complexity of Traversals

What is the complexity of each traversal type?

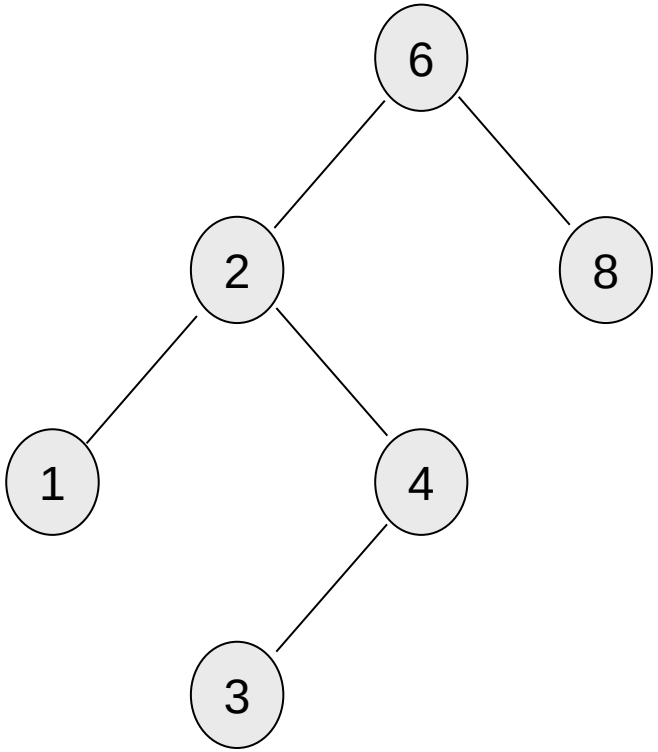
- Preorder traversal
- Postorder traversal
- Inorder traversal

Binary Search Tree

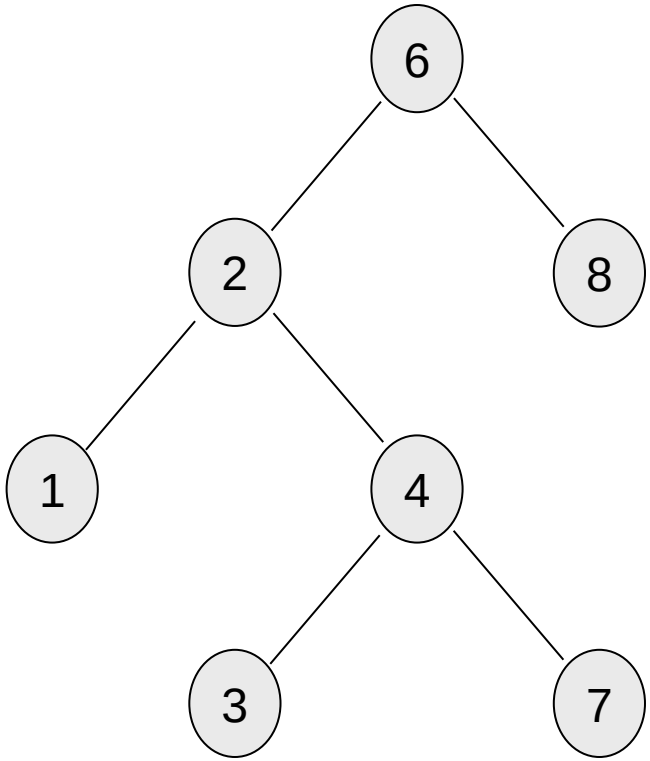
- An important application of binary trees is their use in searching.

- ***Binary search tree*** is a binary tree in which every node X contains a data value that satisfies the following:
  a) all data values in its **left subtree are smaller** than data value in X
  b) all data values in its **right subtree are larger** than data value in X
  c) the left and right subtrees are also binary search trees

Binary Search Tree

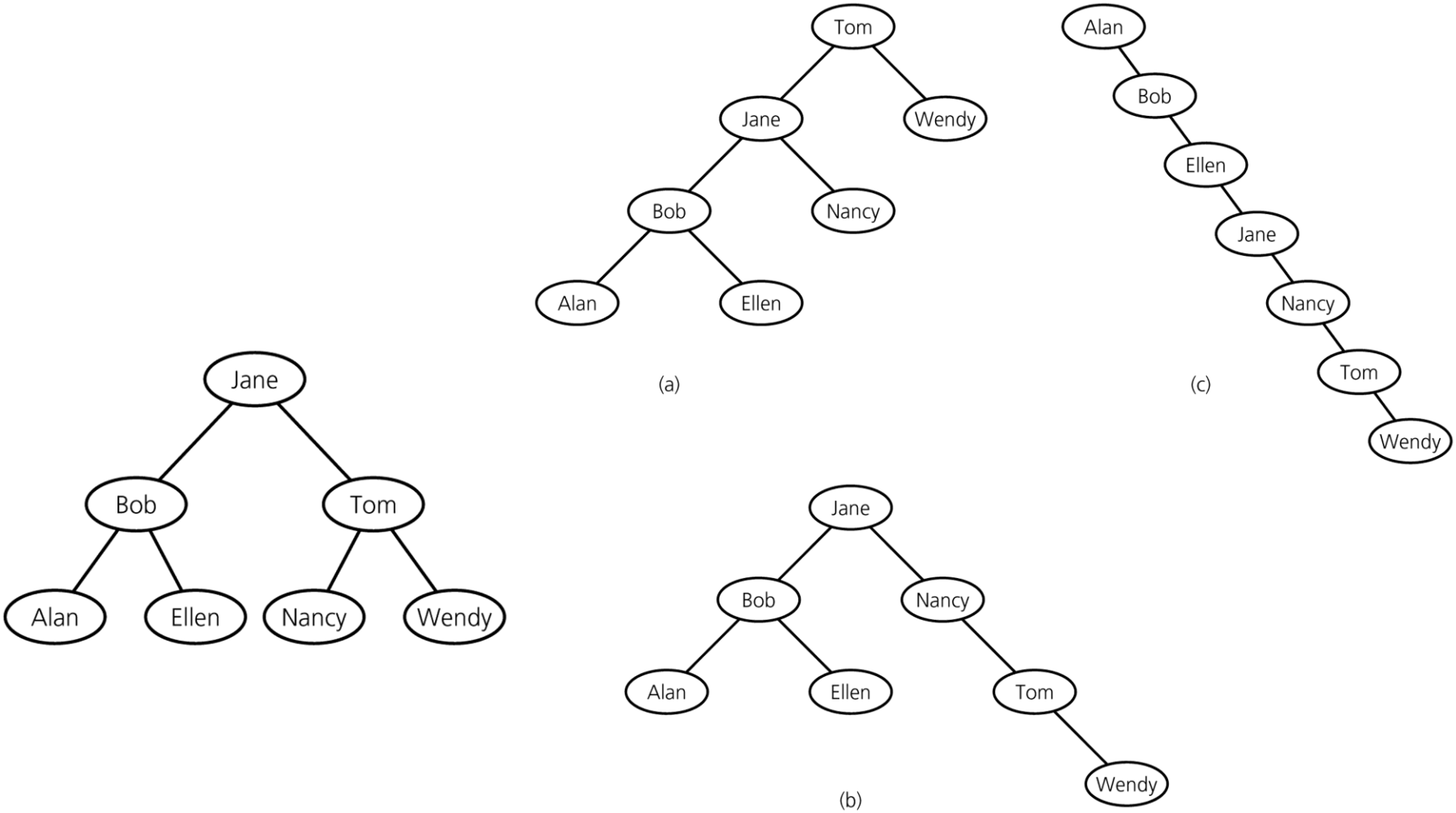


A *binary search tree*



Not a *binary search tree*,
but a *binary tree* **Why?**

Binary Search Trees – containing same data



BinarySearchTree Class – UML Diagram

BinarySearchTree
root *left subtree* *right subtree*
createBinarySearchTree() *destroyBinarySearchTree()* *isEmpty()* *searchTreeInsert()* *searchTreeDelete()* *searchTreeRetrieve()* *preorderTraverse()* *inorderTraverse()* *postorderTraverse()*

The KeyedItem Class

```
typedef desired-type-of-search-key KeyType;

class KeyedItem {
public:
        KeyedItem() { }
        KeyedItem(const KeyType& keyValue) : searchKey(keyValue) { }

        KeyType getKey() const {
                return searchKey;
        }

private:
        KeyType searchKey;
        // ... and other data items
};
```

The TreeNode Class

```
typedef KeyedItem TreeItemType;

class TreeNode {     // a node in the tree
private:
        TreeNode() { }
        TreeNode(const TreeItemType& nodeItem,TreeNode *left = NULL,
                                                        TreeNode *right = NULL)
        : item(nodeItem), leftChildPtr(left), rightChildPtr(right){ }


        TreeItemType item;                  // a data item in the tree
        TreeNode *leftChildPtr;         // pointers to children
        TreeNode *rightChildPtr;


    // friend class - can access private parts
    friend class BinarySearchTree;
};
```

The BinarySearchTree Class

- **Properties**
  - `TreeNode * root`

- **Constructors**
  - `BinarySearchTree();`
  - `BinarySearchTree(const BinarySearchTree& tree);`

- **Destructor**
  - `~BinarySearchTree();`

The BinarySearchTree Class

- **Public methods**
  - bool isEmpty() const;
  - void searchTreeRetrieve(KeyType searchKey, TreeItemType& item);
  - void searchTreeInsert(const TreeItemType& newItem);
  - void searchTreeDelete(KeyType searchKey);
  - void preorderTraverse(FunctionType visit);
  - void inorderTraverse(FunctionType visit);
  - void postorderTraverse(FunctionType visit);
  - BinarySearchTree& operator=(const BinarySearchTree& rhs);

The BinarySearchTree Class

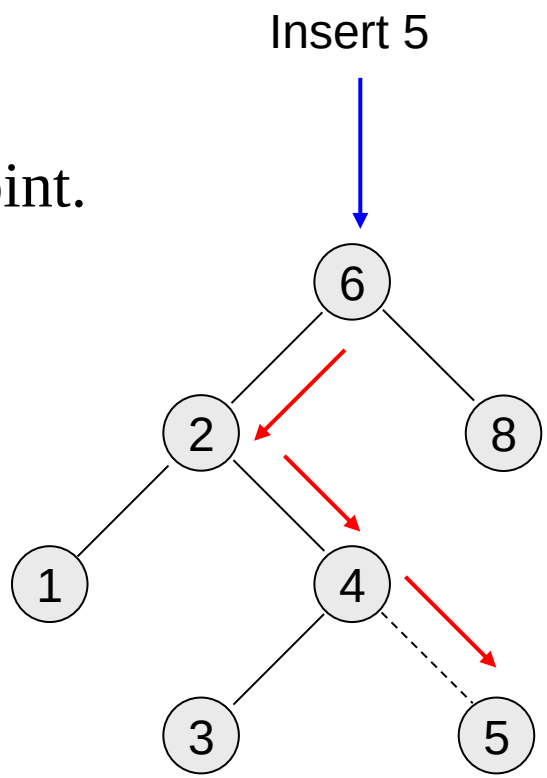
- **`Protected methods`**
  - void retrieveItem(TreeNode *treePtr, KeyType searchKey, TreeItemType& item);
  - void insertItem(TreeNode * &treePtr, const TreeItemType& item);
  - void deleteItem(TreeNode * &treePtr, KeyType searchKey);
  - void deleteNodeItem(TreeNode * &nodePtr);
  - void processLeftmost(TreeNode * &nodePtr, TreeItemType& item);

Searching (Retrieving) an Item in a BST

```
void BinarySearchTree::searchTreeRetrieve(KeyType searchKey,
                                          TreeItemType& treeItem) const throw(TreeException) {
        retrieveItem(root, searchKey, treeItem);
}


void BinarySearchTree::retrieveItem(TreeNode *treePtr, KeyType searchKey,
                                          TreeItemType& treeItem) const throw(TreeException) {

        if (treePtr == NULL)
                throw TreeException("TreeException: searchKey not found");
        else if (searchKey == treePtr->item.getKey())
                treeItem = treePtr->item;
        else if (searchKey < treePtr->item.getKey())
                retrieveItem(treePtr->leftChildPtr, searchKey, treeItem);
        else
                retrieveItem(treePtr->rightChildPtr, searchKey, treeItem);
}
```

Inserting an Item into a BST

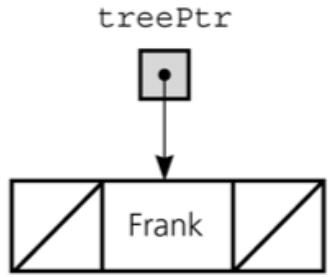
Search determines the insertion point.



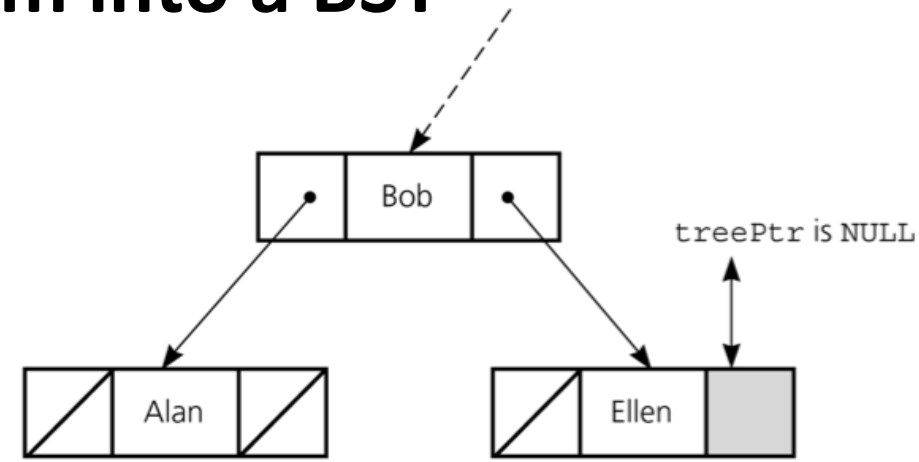
Inserting an Item into a BST

```
void BinarySearchTree::searchTreeInsert(const TreeItemType& newItem) {
        insertItem(root, newItem);
}


void BinarySearchTree::insertItem(TreeNode *& treePtr,
                                const TreeItemType& newItem) throw(TreeException) {

        // Position of insertion found; insert after leaf
        if (treePtr == NULL) {
                treePtr = new TreeNode(newItem, NULL, NULL);
                if (treePtr == NULL)
                        throw TreeException("TreeException: insert failed");
        }
        // Else search for the insertion position
        else if (newItem.getKey() < treePtr->item.getKey())
                insertItem(treePtr->leftChildPtr, newItem);
        else
                insertItem(treePtr->rightChildPtr, newItem);
}
```

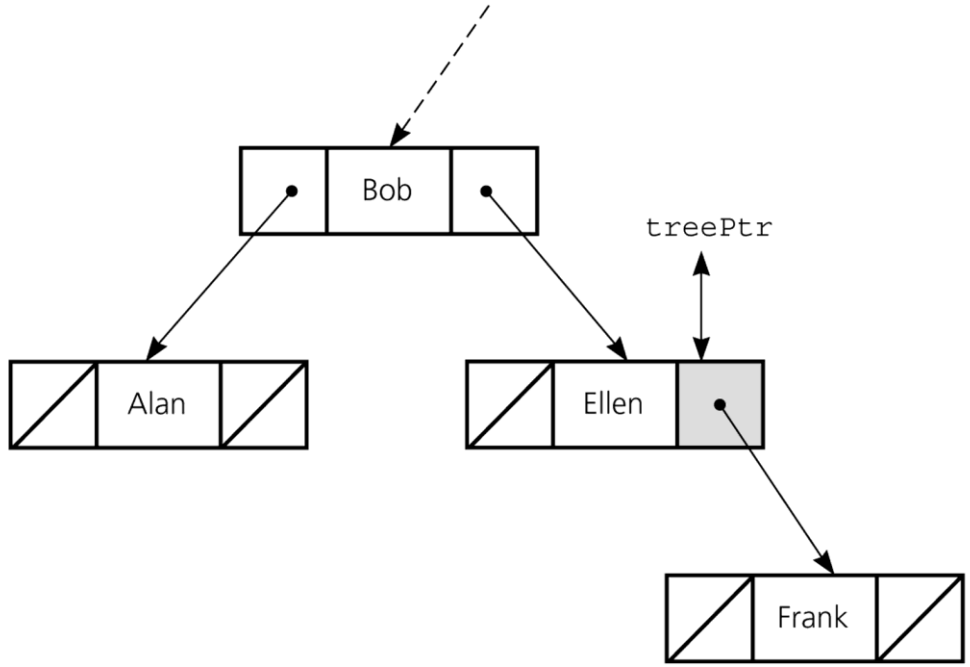
Inserting an Item into a BST



(a)



(b)



(c)

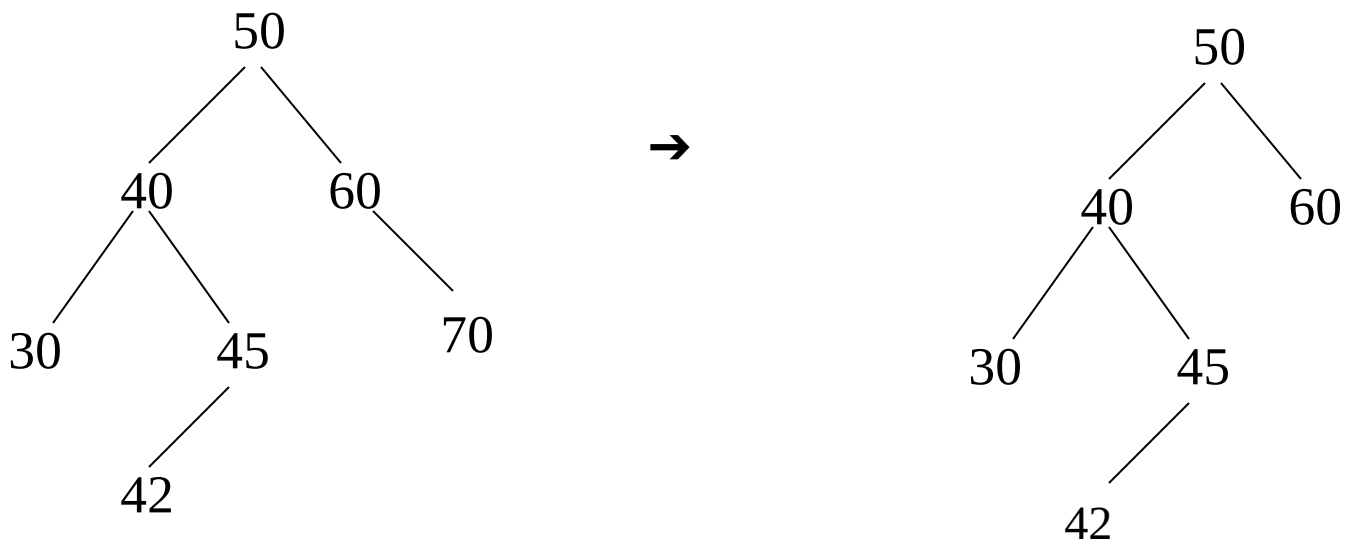
Deleting An Item from a BST

- To delete an item from a BST, we have to locate that item in that BST.

- The deleted node can be:
  - *Case 1* – A leaf node.
  - *Case 2* – A node with only one child
    (with left child or with right child).
  - *Case 3* – A node with two children.

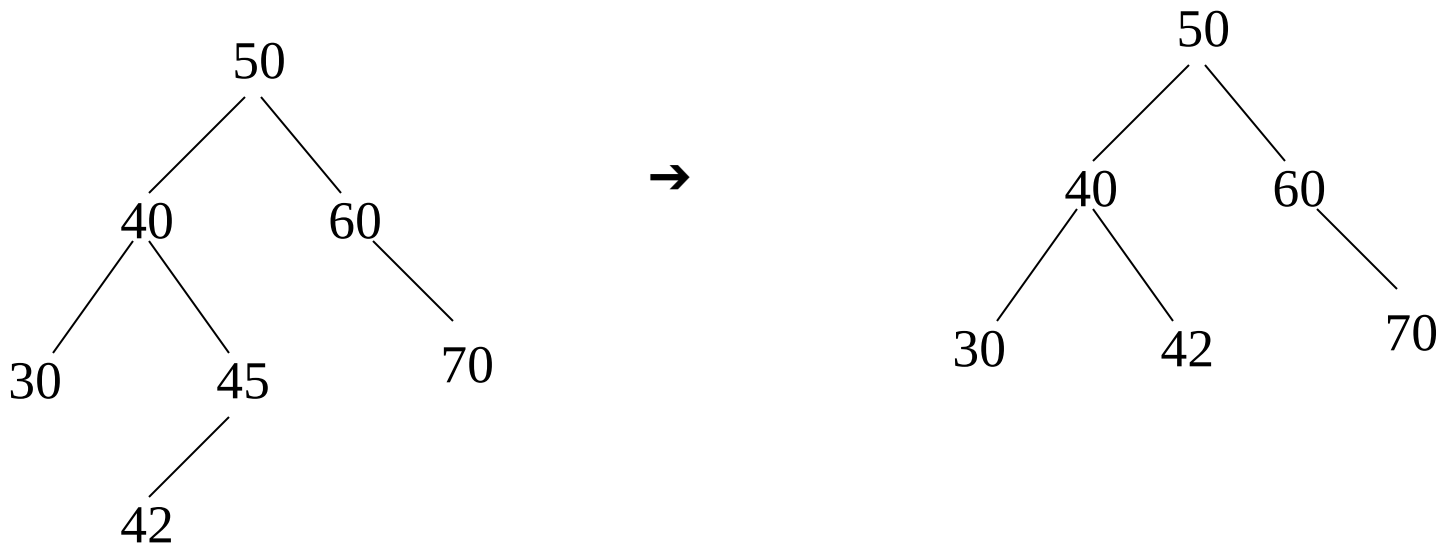
Deletion – Case 1: A Leaf Node

To remove the leaf containing the item, we have to set the pointer in its parent to NULL.



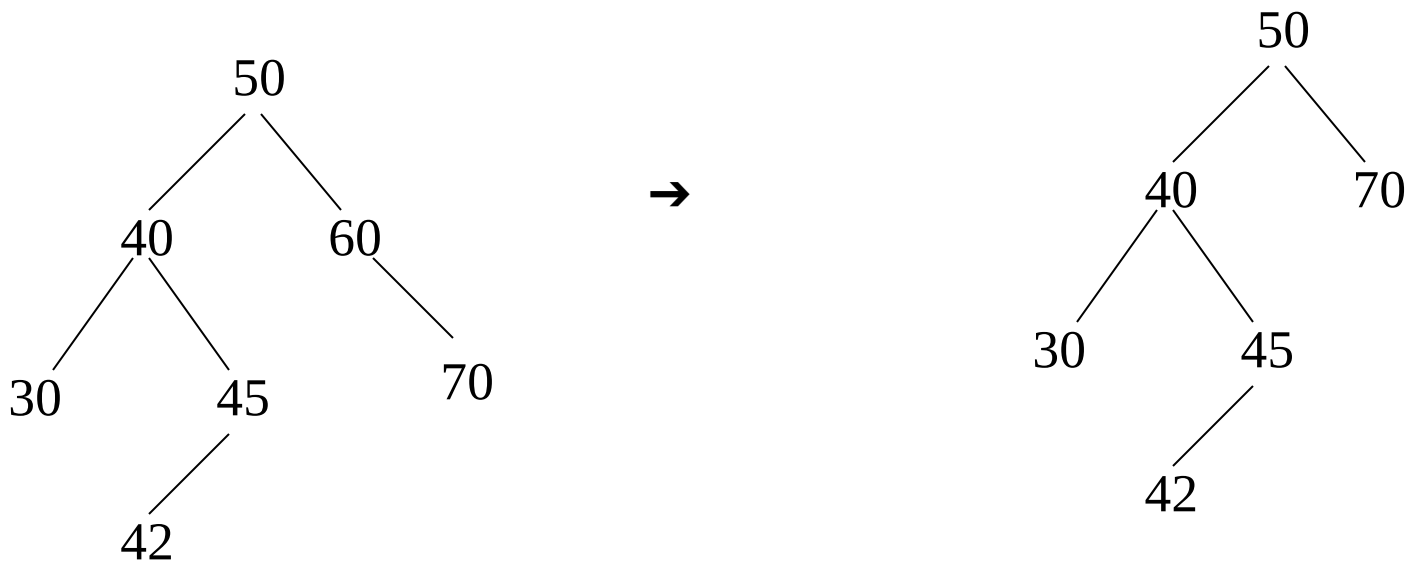
Delete 70 (A leaf node)

Deletion – Case 2: A Node with only a left child



Delete 45 (A node with only a left child)

Deletion – Case 2: A Node with only a right child



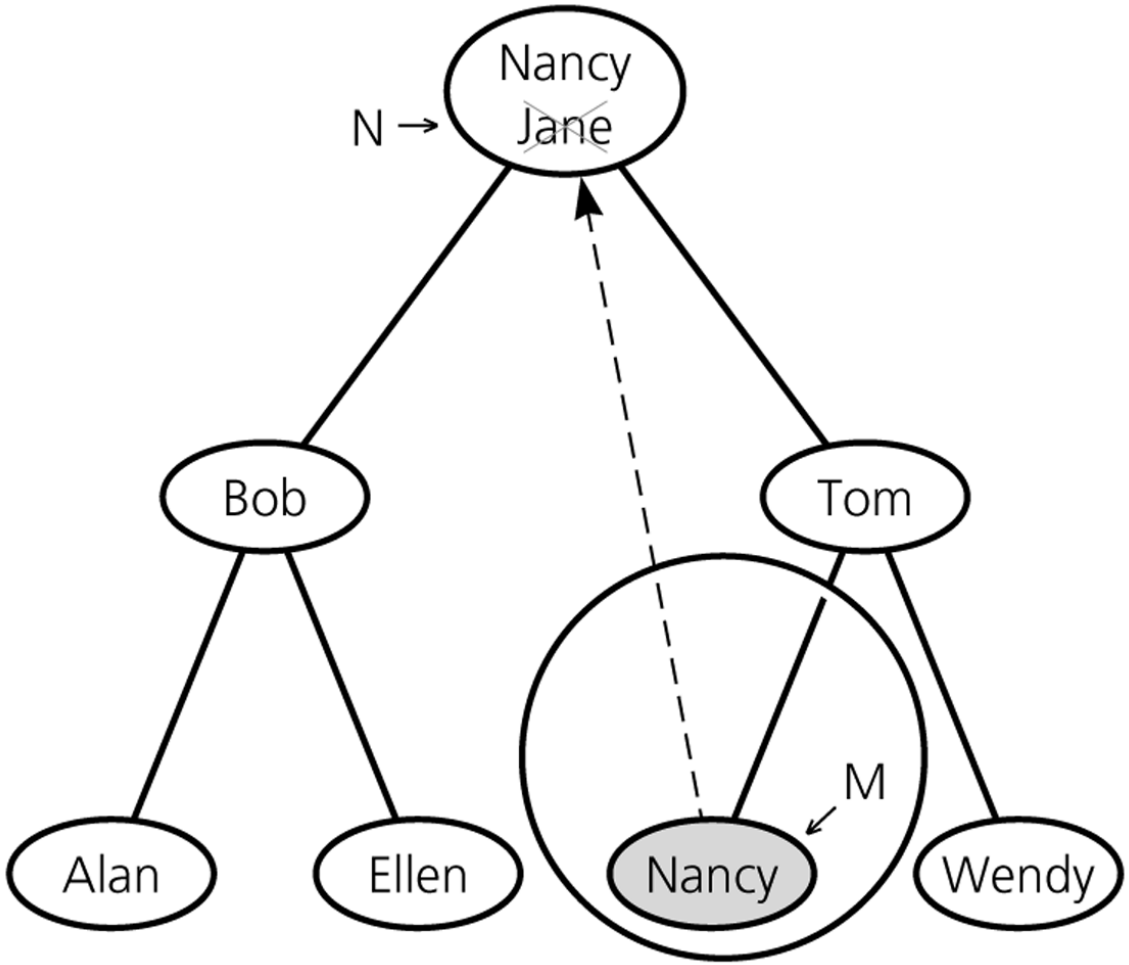
Delete 60 (A node with only a right child)

Deletion – Case 3: A Node with two children

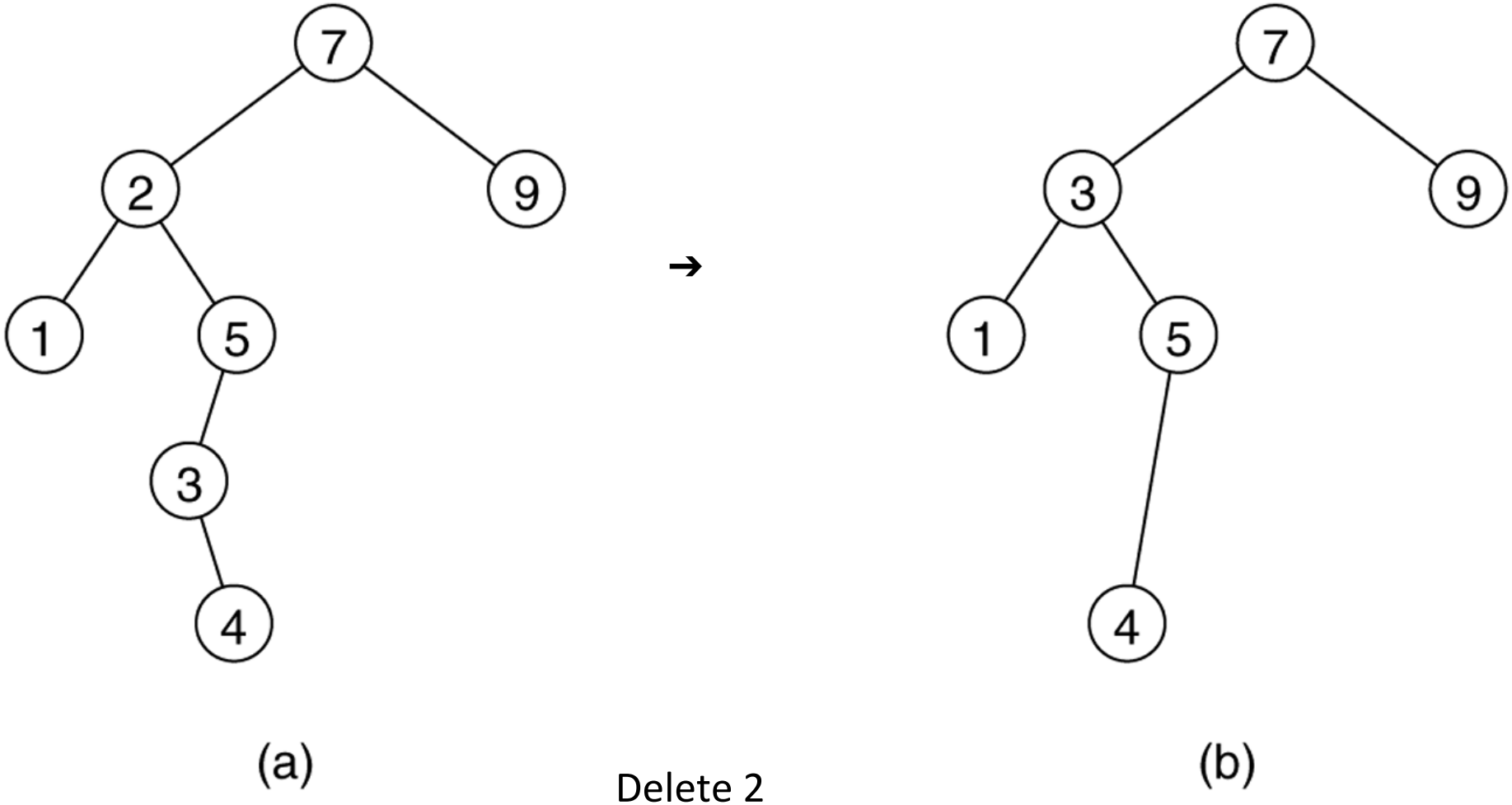
- **Locate** the inorder successor of the node.
- **Copy** the item in this node into the node which contains the item which will be deleted.
- **Delete** the node of the inorder successor.

Delete 40 (A node with two children)

Deletion – Case 3: A Node with two children



Deletion – Case 3: A Node with two children



Delete 2

Deletion from a BST

```
void BinarySearchTree::searchTreeDelete(KeyType searchKey)  throw(TreeException) {
        deleteItem(root, searchKey);
}

void BinarySearchTree::deleteItem(TreeNode * &treePtr, KeyType searchKey)  throw(TreeException) {
        if (treePtr == NULL) // Empty tree
                throw TreeException("Delete failed");

        // Position of deletion found
        else if (searchKey == treePtr->item.getKey())
                deleteNodeItem(treePtr);

        // Else search for the deletion position
        else if (searchKey < treePtr->item.getKey())
                deleteItem(treePtr->leftChildPtr, searchKey);
        else
                deleteItem(treePtr->rightChildPtr, searchKey);
}
```

Deletion from a BST

```
void BinarySearchTree::deleteNodeItem(TreeNode * &nodePtr) {
        TreeNode *delPtr;
        TreeItemType replacementItem;


        // (1)  Test for a leaf
        if ( (nodePtr->leftChildPtr == NULL) &&
           (nodePtr->rightChildPtr == NULL) ) {
                delete nodePtr;
                nodePtr = NULL;
        }


        // (2)  Test for no left child
        else if (nodePtr->leftChildPtr == NULL){
                delPtr = nodePtr;
                nodePtr = nodePtr->rightChildPtr;
                delPtr->rightChildPtr = NULL;
                delete delPtr;
        }
```

Deletion from a BST

```
        // (3)  Test for no right child
        else if (nodePtr->rightChildPtr == NULL) {
                // ...
                // Left as an exercise
        }


        // (4)  There are two children:
        //      Retrieve and delete the inorder successor
        else {
                processLeftmost(nodePtr->rightChildPtr, replacementItem);
                nodePtr->item = replacementItem;
        }

}
```

Deletion from a BST

```
void BinarySearchTree::processLeftmost(TreeNode *&nodePtr,
                                                    TreeItemType& treeItem){

        if (nodePtr->leftChildPtr == NULL) {
                treeItem = nodePtr->item;
                TreeNode *delPtr = nodePtr;
                nodePtr = nodePtr->rightChildPtr;
                delPtr->rightChildPtr = NULL; // defense
                delete delPtr;
        }
        else
                processLeftmost(nodePtr->leftChildPtr, treeItem);
}
```

Traversals

- The traversals for binary search trees are same as the traversals for the binary trees.

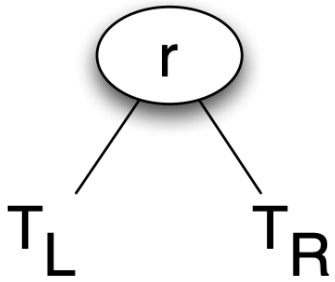
Theorem: Inorder traversal of a binary search tree will visit its nodes in sorted search-key order.

Proof: Proof by induction on the height of the binary search tree T.

Basis: h=0 ➔ no nodes are visited, empty list is in sorted order.

Inductive Hypothesis: Assume that the theorem is true for all k, $0 \leq k < h$

Inductive Conclusion: We have to show that the theorem is true for k=h>0. T should be:



Since the lengths of T_L and T_R are less than h, the theorem holds for them. All the keys in T_L are less than r, and all the keys in T_R are greater than r. In inorder traversal, we visit T_L first, then r, and then T_R. Thus, the theorem holds for T with height k=h.

Minimum Height

- **Complete trees and full trees have minimum height.**
- The height of an n-node binary search tree ranges from $\lceil \log_2(n+1) \rceil$ to n.

- **Insertion in search-key order** produces a maximum-height BST.
- **Insertion in random order** produces a near-minimum-height BST.

- That is, the height of an n-node binary search tree
  - *Best Case* – $\lceil \log_2(n+1) \rceil$ → $O(\log_2 n)$
  - *Worst Case* – n → $O(n)$
  - *Average Case* – close to $\lceil \log_2(n+1) \rceil$ → $O(\log_2 n)$

Average Height

- If we insert n items into an empty BST to create a BST with n nodes,
  - How many different binary search trees with n nodes?
  - What are their probabilities?

- There are n! different orderings of n keys.
  - But how many different binary search trees with n nodes?

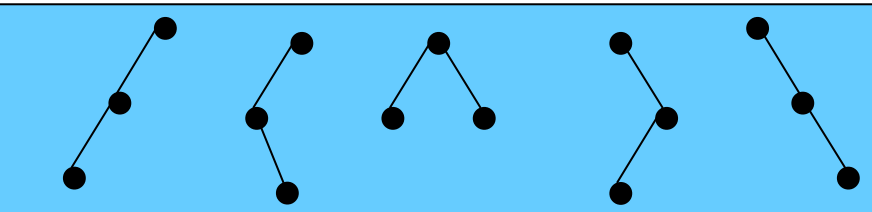
n=0 ➔ 1 BST (empty tree)

n=1 ➔ 1 BST (a binary tree with a single node)

n=2 ➔ 2 BSTs

n=3 ➔ 5 BSTs

<u>When n=3</u>



Probabilities:	1/6	1/6	2/6	1/6	1/6
Insertion Order:	3,2,1	3,1,2	2,1,3 2,3,1	1,3,2	1,2,3

Order of Operations on BSTs

Operation	Average case	Worst case
Retrieval	O(log n)	O(n)
Insertion	O(log n)	O(n)
Deletion	O(log n)	O(n)
Traversal	O(n)	O(n)

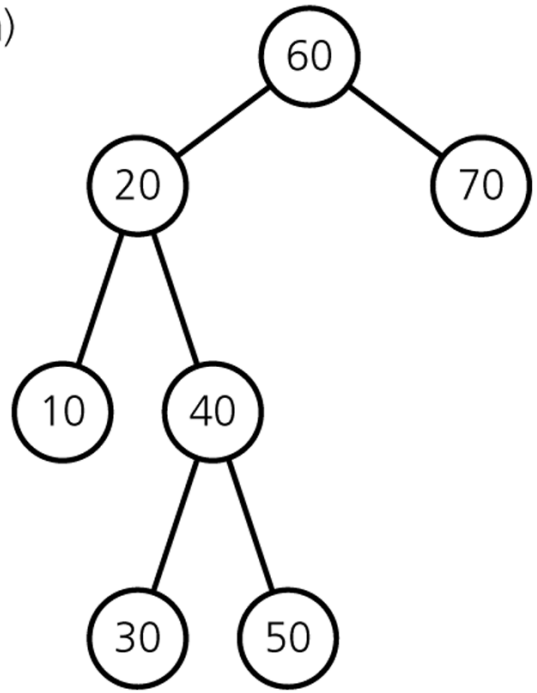
Saving a BST into a file and restoring it to its original shape

- **Save:**
  - Use a preorder traversal to save the nodes of the BST into a file

- **Restore:**
  - Start with an empty BST
  - Read the nodes from the file one by one and insert them into the BST

Saving a BST into a file and restoring it to its original shape

(a)



(b)

```
bst.searchTreeInsert(60);
bst.searchTreeInsert(20);
bst.searchTreeInsert(10);
bst.searchTreeInsert(40);
bst.searchTreeInsert(30);
bst.searchTreeInsert(50);
bst.searchTreeInsert(70);
```

Preorder: 60 20 10 40 30 50 70

Saving a BST into a file and restoring it to a minimum-height BST

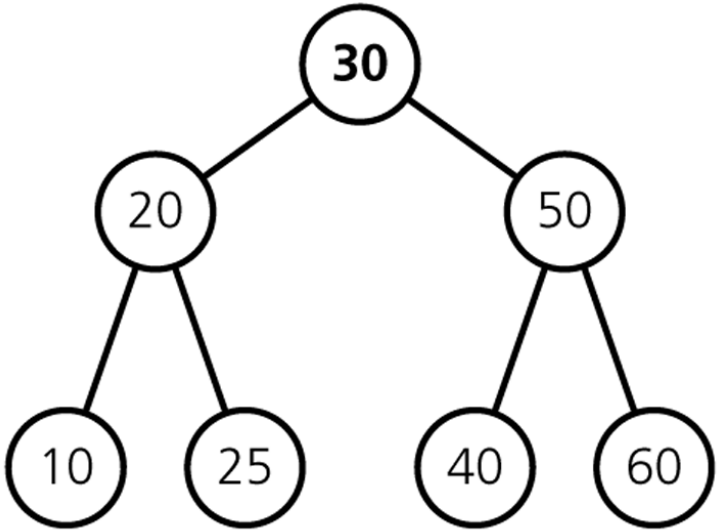
- **Save:**
  - Use an inorder traversal to save the nodes of the BST into a file. The saved nodes will be in ascending order
  - Save the number of nodes (n) in somewhere

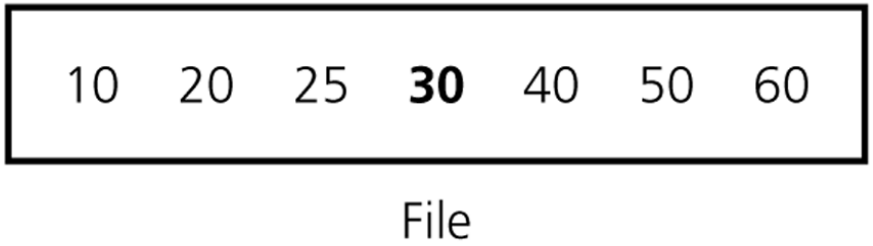
- **Restore:**
  - Read the number of nodes (n)
  - Start with an empty BST
  - Read the nodes from the file one by one to create a minimum-height binary search tree

Building a minimum-height BST

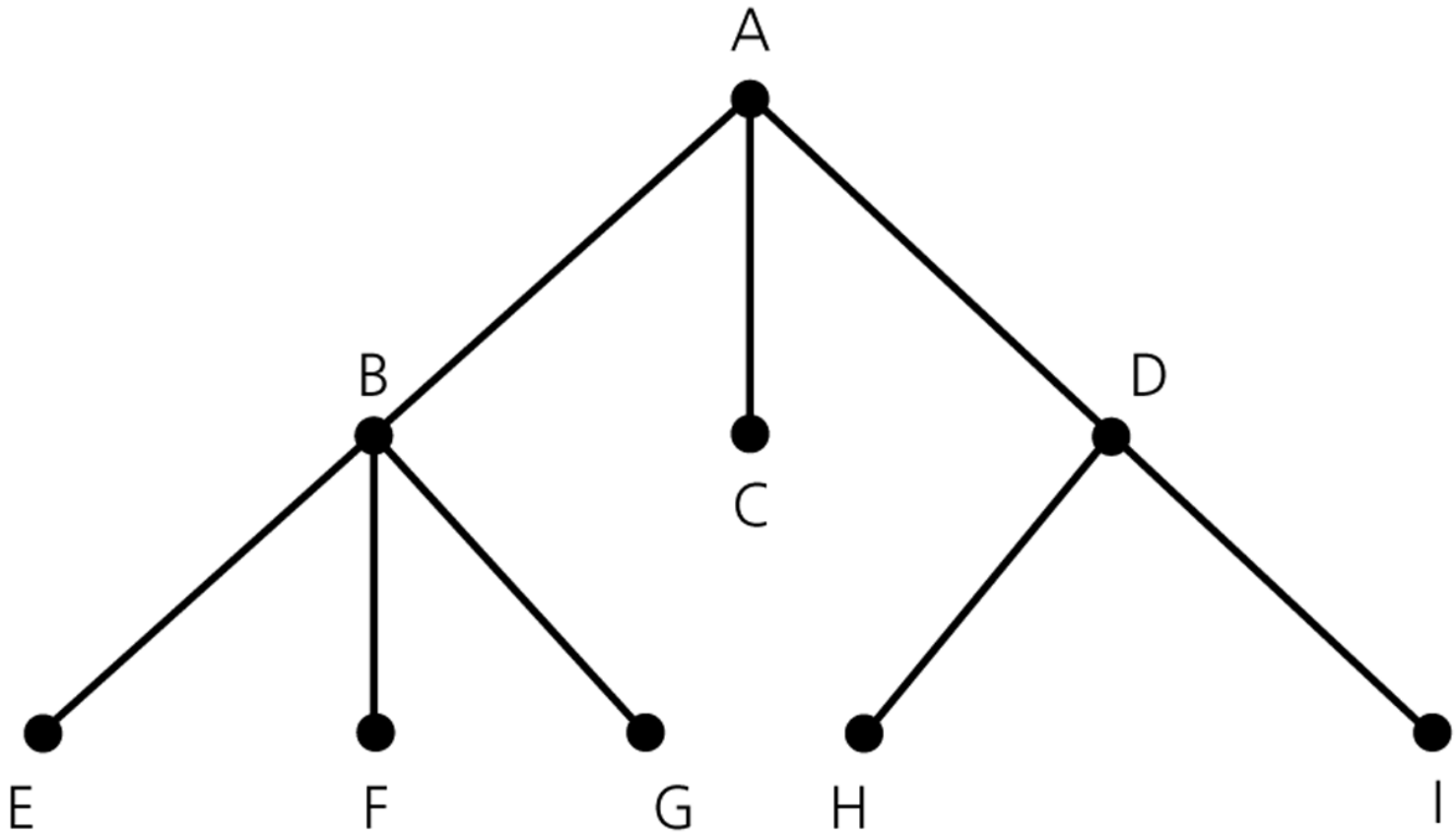
```
// Builds a minimum-height binary search tree from n sorted
// values in a file. treePtr will point to the tree's root.
readTree(out treePtr:TreeNodePtr, in n:integer)

        if (n>0) {
                treePtr = pointer to new node with NULL child pointers

                // construct the left subtree
                readTree(treePtr->leftChildPtr, n/2)

                // get the root
                Read item from file into treePtr->item

                // construct the right subtree
                readTree(treePtr->rightChildPtr, (n-1)/2)
        }
```

A full tree saved in a file by using inorder traversal

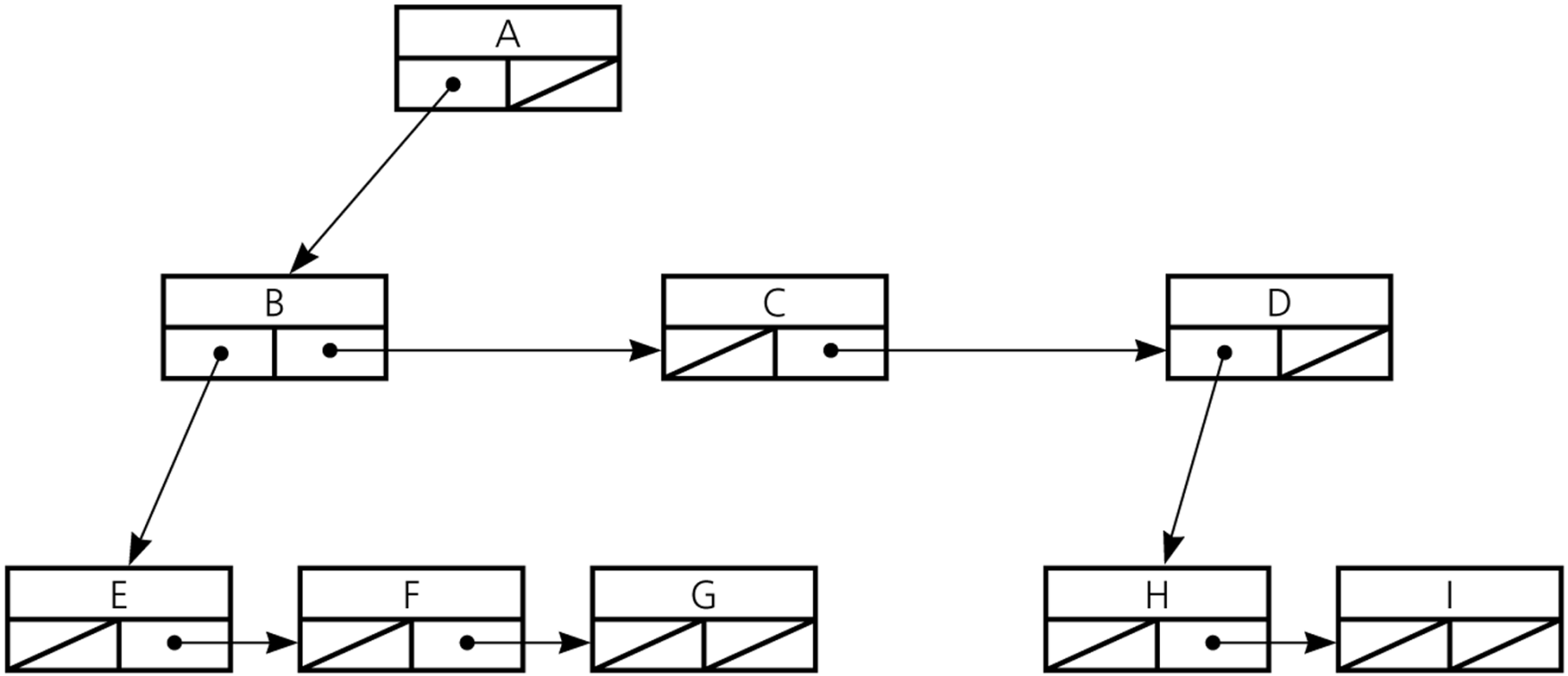




A General Tree

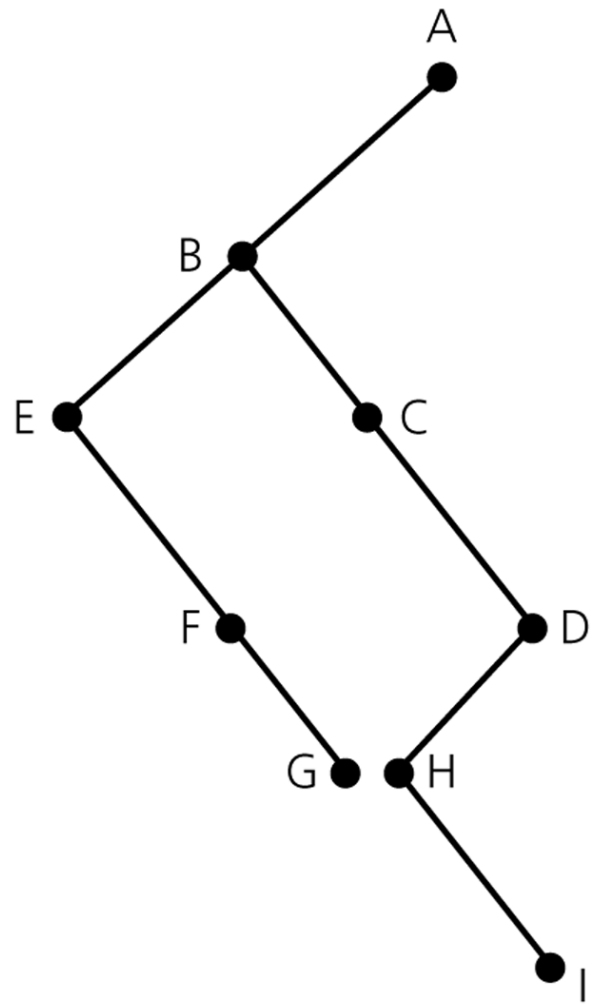


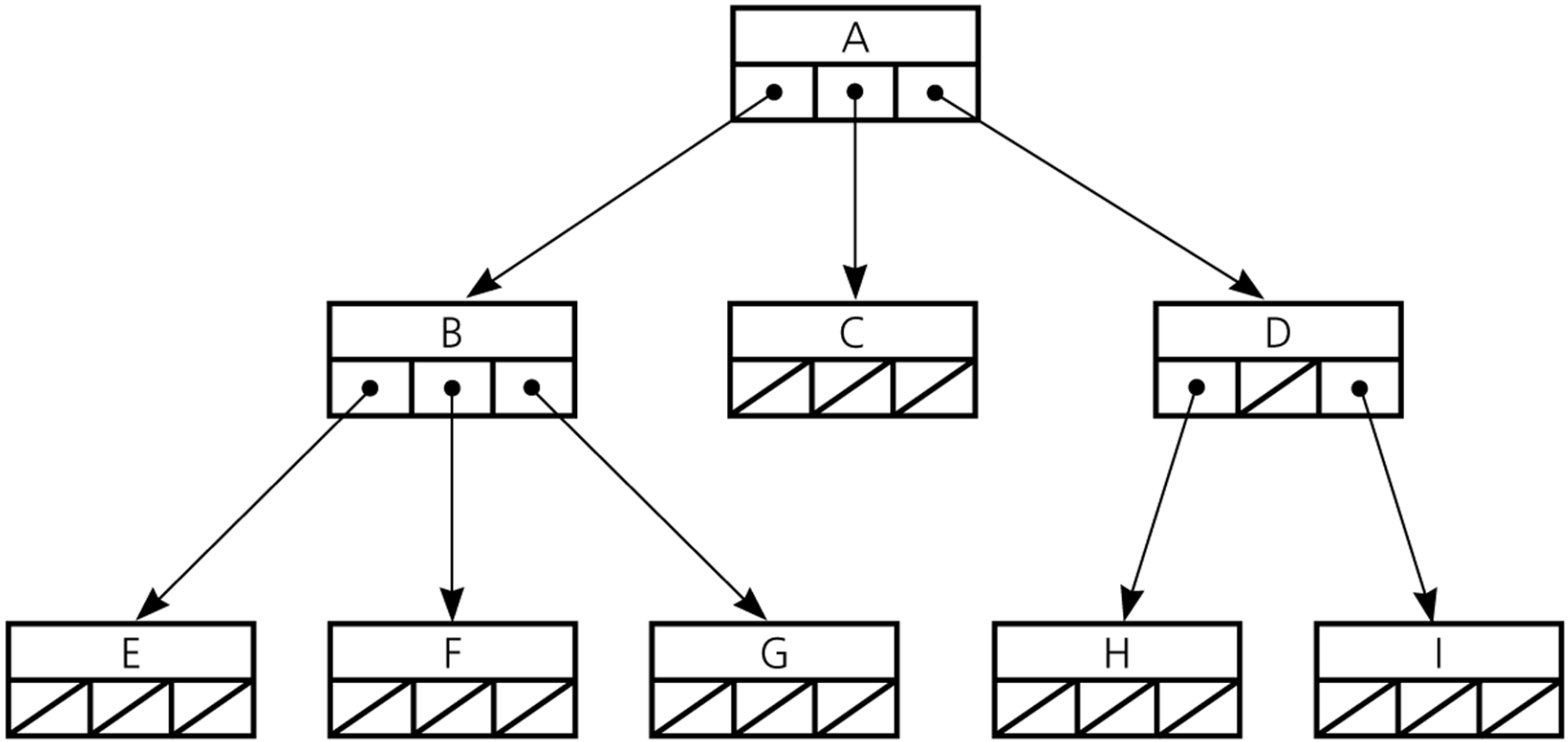
A Pointer-Based Implementation of General Trees



A Pointer-Based Implementation of General Trees

A pointer-based implementation of a general tree can also represent a binary tree.



N-ary Tree

An **n-ary tree** is a generalization of a binary whose nodes each can have no more than n children.